Catalog No. C10108B

DADCO®

Full and Compact Height Nitrogen Gas Springs 90.10 / 90.8 Series



90.10 Series
Full Height /
Matches ISO Standards

90.8 Series
Compact Height
Alternative

PED
97/23/EC
COMPLIANT

Introduction

DADCO®

The global leader in nitrogen gas spring technology

Introduction

DADCO produces top quality products at competitive prices and provides a superior level of customer service. Founded in 1958, DADCO is the highest volume producer of gas springs for press tools. DADCO's products are widely approved and used in global operations for many industries including metal stamping, automotive and plastic injection molding.



90.10 and 90.8 Series

DADCO's full height 90.10 Series matches the ISO Standard specifications and offers the widest choice of sizes, stroke lengths and accessories available. All international automotive standard mounts (welded and attachable) are offered, as well as legacy mounts for replacement of older springs, see pages 4-18.

Using technology from the popular 90.9 Series, DADCO's compact 90.8 Series is 25 mm – 50 mm *(0.98˝ – 1.97˝)* shorter than ISO Standard Nitrogen Gas Springs. It is available in several models with stroke lengths up to 250 mm.

Can Diameter (mm)	Model	90.10 Series Overall Length (mm)	90.8 Series Overall Length (mm)	Maximum Force on Contact
44.5	00500	2 x Stroke + 85	N/A	5 kN
50	00750	2 x Stroke + 95	2 x Stroke + 70	7.5 kN
75	01500	2 x Stroke + 110	2 x Stroke + 85	15 kN
95	03000	2 x Stroke + 120	2 x Stroke + 95	30 kN
120	05000	2 x Stroke + 140	2 x Stroke + 102.5	50 kN
150	07500	2 x Stroke +155	2 x Stroke +105	75 kN
195	10000	2 x Stroke +160	N/A	100 kN

High Quality Construction

DADCO's 90.10 and 90.8 Series Gas Springs feature: a one-piece body (base-to-tube welding process verified ultrasonically); impact fatigue-resistant alloy steel super finished piston rod; rod cartridge with a proprietary double-lip rod seal; low friction rod bearings; a glass-filled, nylon guide ring on the piston retainer and special high temperature synthetic lubricant. The result of this combination is a superior design that ensures low friction, sealability and excellent wear characteristics essential for optimum gas spring life.

Customer Satisfaction

DADCO's motto is "*Whatever It Takes To Satisfy Our Customers*." DADCO will assist in any way possible to ensure that customers are completely satisfied. DADCO's salespeople and distributors are solution-oriented, product-knowledgeable and eager to assist customers. DADCO's engineers are available to help customers with specific applications.

Numerous Piping Options

With DADCO's 90.10 and 90.8 Series, piping gas springs is easy because every necessary tool or component is readily available. For more information, refer to DADCO's Nitrogen Gas Spring Linked System Components Catalog. Training and technical assistance are also offered by DADCO.



SMS®

For those instances when a customer prefers to have DADCO provide a finished system, several options are available. A popular option, DADCO's Sectional Mounting System (SMS®) includes cylinders mounted to a SMS® plate and linked using hose, fittings and a control panel. Shipped ready to install, customers find the SMS® to be a trouble free, cost-effective option. For additional information, refer to catalog C13106.

SMS-i®

Another option is the Sectional Mounting System-Internal (SMS-i®). The SMS-i® consists of an internally piped plate with mounted cylinders. DADCO recommends using the SMS-i® as a cost- effective alternative to traditional manifold systems. For more information, request catalog C13106.

Guaranteed Long Life

In factory testing and field experience, the service life of DADCO's 90.10 and 90.8 Series Gas Springs consistently exceeds one million strokes. This is supported by DADCO's written One Year/One Million Stroke ***Gold Guarantee***. Contact DADCO for more information.

Warranty

DADCO warrants its products to be free from defects in workmanship or materials for a period of one year from date of manufacture.

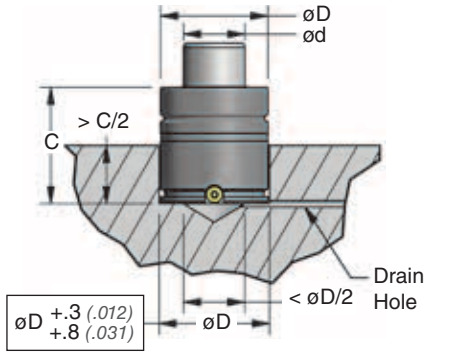
CAD Templates

DADCO's entire product line is available on-line in solid models and 2D CAD formats. For more information, visit our website, www.dadco.net, or contact DADCO.

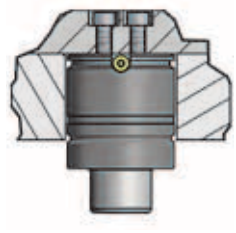
734.207.1100 • 800.DADCO.USA • fax 734.207.2222 • www.dadco.net

Application Examples

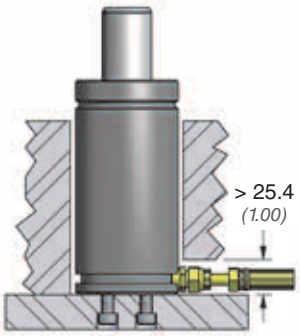
DADCO offers a variety of mount options to meet specific customer applications. Installation and fastening of the gas springs should take into consideration load support, fastener selection and torque values. For additional information on installation requirements see page 23. Cylinder and mount dimensions are shown on pages 4-18.



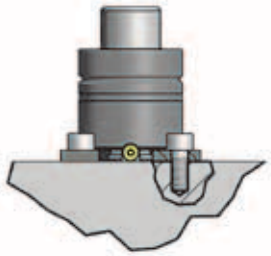
TO Basic Model in a flat bottom pocket. The pocket must be bored with a flat bottom, or a spacer must be used to create a flat surface.



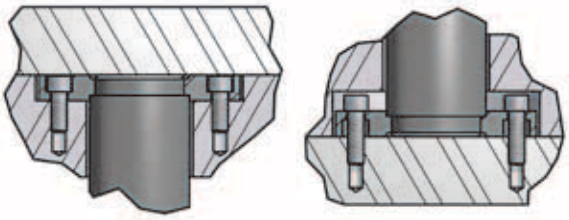
TO Basic Model mounted inverted requires back-up to support the full load. Retain inverted cylinders tight in the pocket with the appropriate length cap screw to eliminate movement.



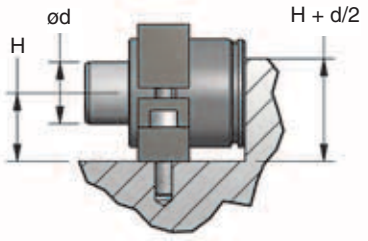
TO Basic Model mounted to a plate. Linked cylinders require clearance for the hose and fittings.



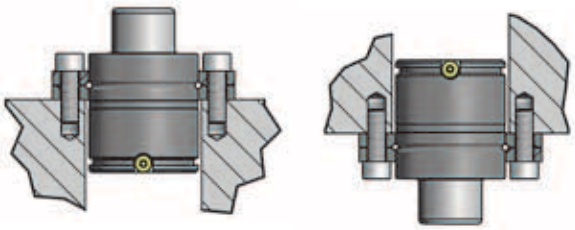
B12/B312 mounts must be fastened to the bottom groove only. Back-up is required to support the full load.



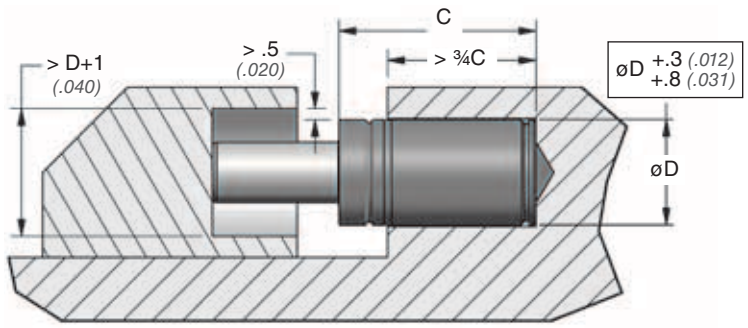
TFM/TSW/B11 mounts may be attached from the top or base. Back-up is required to support the full load. *NOTE: Mount 90.11.07500 from top only.*



B19/B319V/B26 mounts require back-up to support the load.

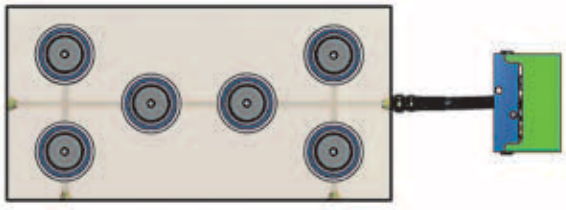


B21/B25 mounts must be fastened to the top groove only. The wire ring supplied with mounts supports the full load.

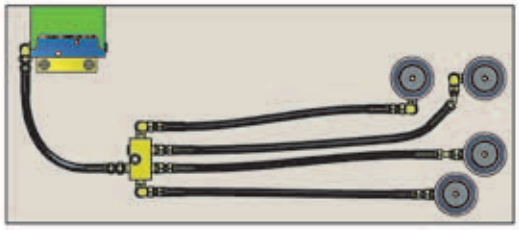


TO Basic Model mounted in a horizontal pocket. Allow clearance in the rod pocket to avoid contacting the body of the cylinder during operation. Allow the rod to locate freely.

SMS® / SMS-i® DADCO offers customized Sectional Mounting Systems (SMS® and SMS-i®) which are fabricated to customer specifications, pressure tested and shipped ready to install.

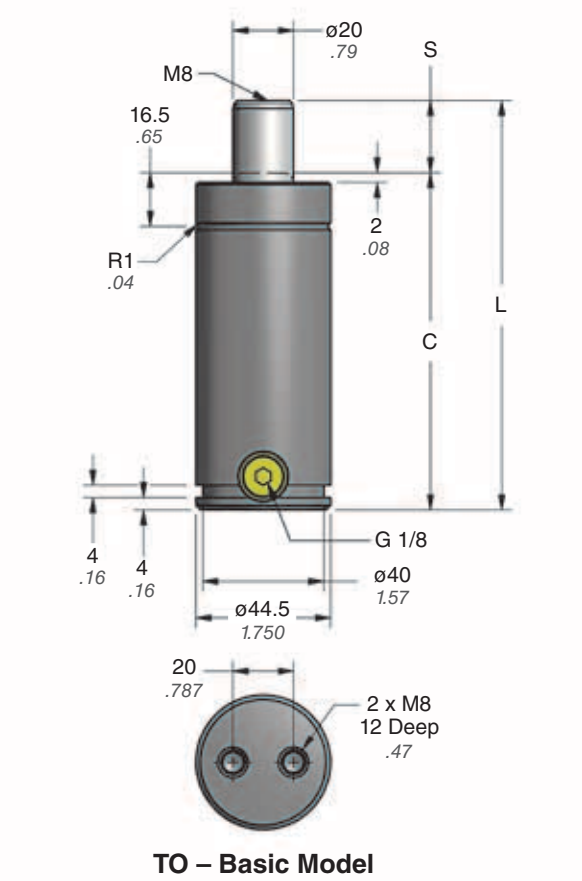


DADCO SMS-i® with (6) 90.10.05000 gas springs plumbed internally and connected to a control panel.



DADCO SMS® with (4) 90.10.03000 gas springs linked using 90.400 *(Y-400)* hose, fittings and a control panel.

5 kN / 0.5 ton



TO – Basic Model

Model Not Available in 90.8 Series

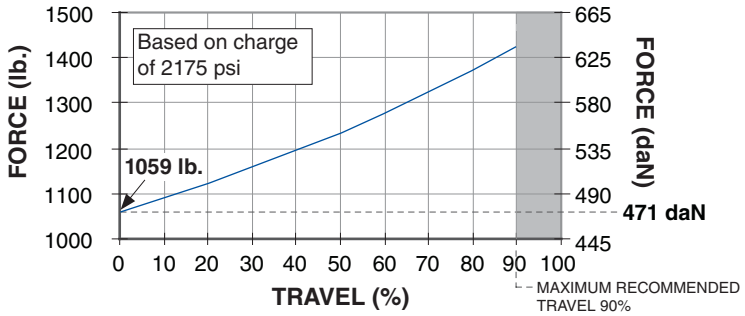
90.10.00500 ISO			
Part No.	**S** mm *inch*	**C**	**L** ±0.25 *±.010*
90.10.00500.013	12.5 *0.49*	97.5 *3.84*	110 *4.331*
• 90.10.00500.025	25 *0.98*	110 *4.33*	135 *5.315*
90.10.00500.038	37.5 *1.48*	122.5 *4.82*	160 *6.299*
• 90.10.00500.050	50 *1.97*	135 *5.31*	185 *7.283*
90.10.00500.063	62.5 *2.46*	147.5 *5.81*	210 *8.268*
• 90.10.00500.080	80 *3.15*	165 *6.50*	245 *9.646*
90.10.00500.100	100 *3.94*	185 *7.28*	285 *11.220*
90.10.00500.125	125 *4.92*	210 *8.27*	335 *13.189*
90.10.00500.160	160 *6.30*	245 *9.65*	405 *15.945*
90.10.00500.200	*200* *7.87*	285 *11.22*	485 *19.094*

• ISO standard stroke lengths

Force Charts

On-Contact Force

psi	lb.
2175	1059
2000	974
1750	852
1500	730
1000	487
500	243
250	122



On-Contact Force

bar	daN
150	471
125	393
100	314
75	236
50	157
25	79
20	63

Ordering Example:

90.10.00500.025. *TO.* *C.* *150*

Part Number: Includes Series (90.10), Model and Stroke Length.

Mount Option: TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

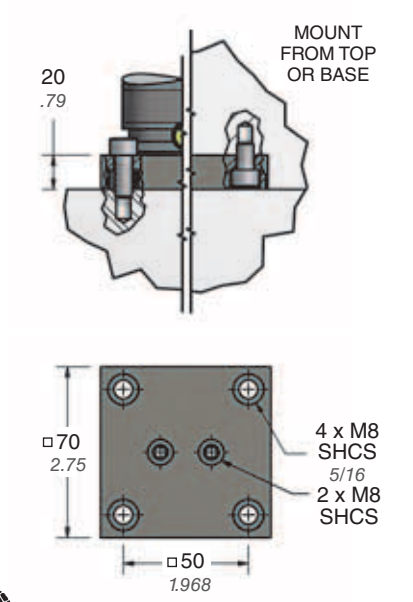
Charging Pressure: 15-150 bar *(220-2175 psi). When not specified, default is 150 bar.*

Operating System: C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

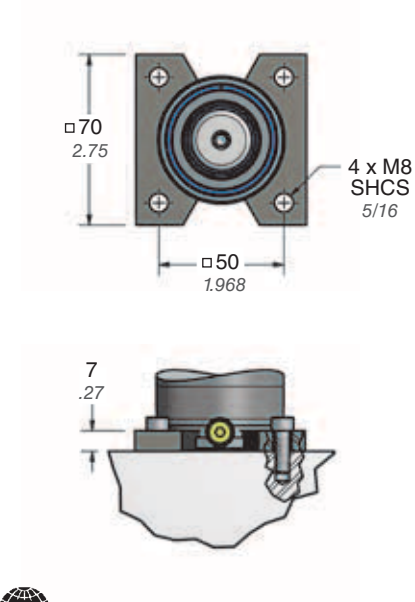
734.207.1100 • 800.DADCO.USA • fax 734.207.2222 • www.dadco.net

5 kN / 0.5 ton

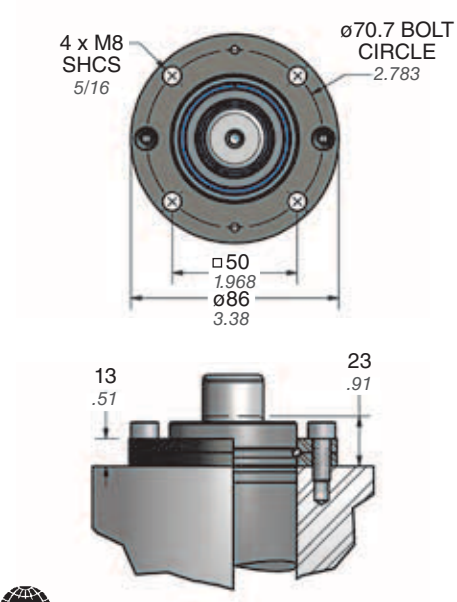
B11 — 90.11.00500



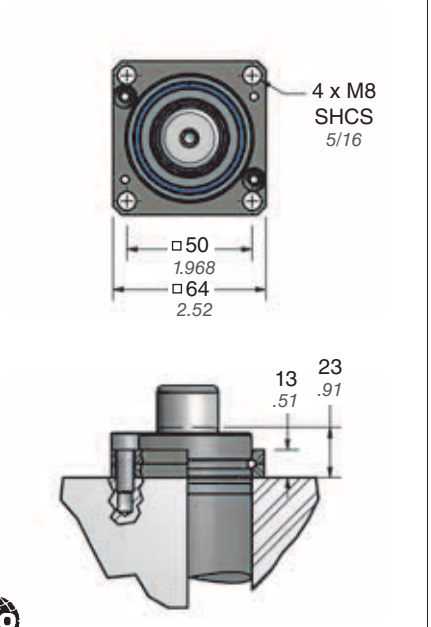
B312 — 90.312.00500



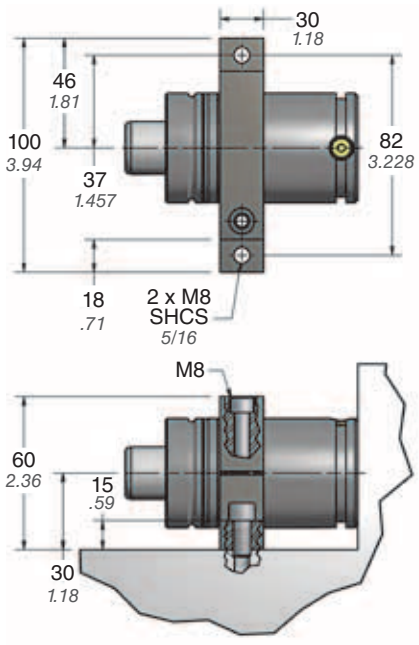
B21 — 90.21.00500



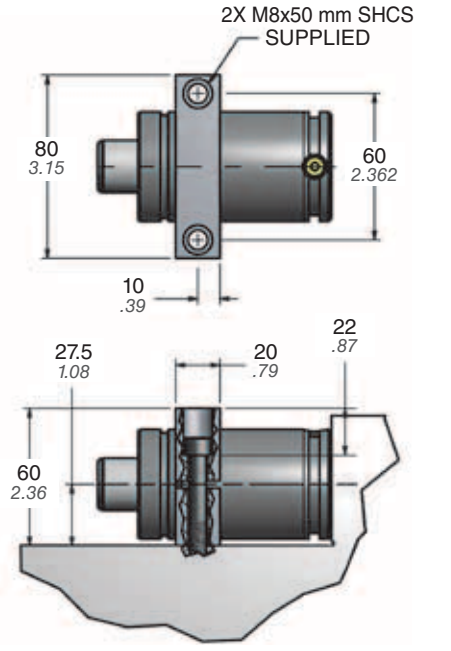
B25 — 90.25.00500



B19 — 90.19.00500



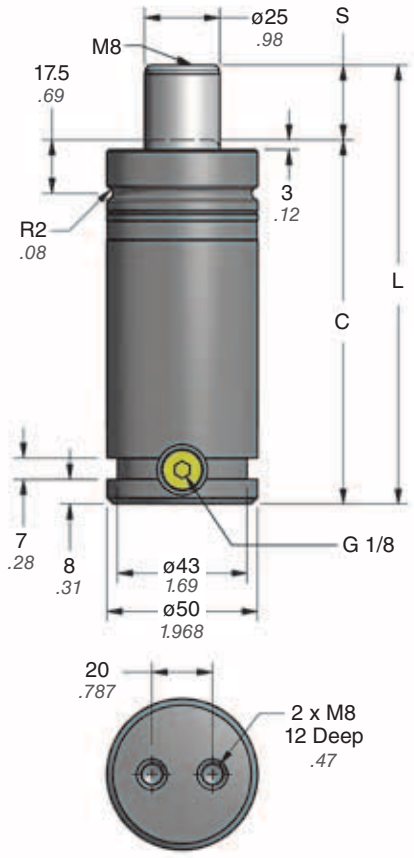
B319V — 90.319V.0800



Ordering Example: **Cylinder with Mount:** 90.10.00500.025.B21.C.150 **Mount Only:** 90.21.00500

90.10 and 90.8 Series Nitrogen Gas Springs

7.5 kN / 0.75 ton



TO – Basic Model

25 mm shorter!

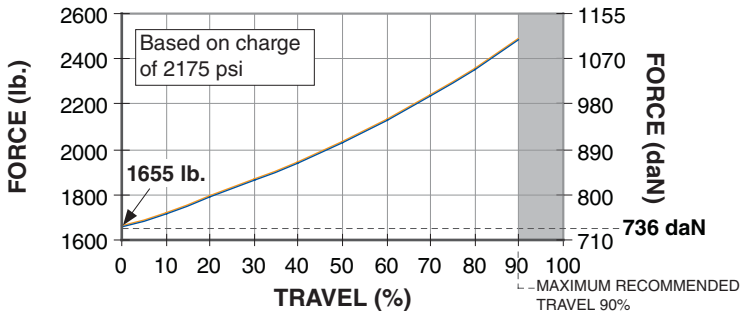
Part No.	S mm *inch*	90.10.00750 ISO C	90.10.00750 ISO L ±0.25 *±.010*	90.8.00750 C	90.8.00750 L ±0.25 *±.010*
90.__.00750.013	12.5 *0.49*	107.5 *4.23*	120 *4.724*	82.5 *3.25*	95 *3.740*
• 90.__.00750.025	25 *0.98*	120 *4.72*	145 *5.709*	95 *3.74*	120 *4.724*
90.__.00750.038	37.5 *1.48*	132.5 *5.22*	170 *6.693*	107.5 *4.23*	145 *5.709*
• 90.__.00750.050	50 *1.97*	145 *5.71*	195 *7.677*	120 *4.72*	170 *6.693*
90.__.00750.063	62.5 *2.46*	157.5 *6.20*	220 *8.661*	132.5 *5.22*	195 *7.677*
90.__.00750.075	75 *2.95*	170 *6.69*	245 *9.646*	145 *5.71*	220 *8.661*
• 90.__.00750.080	80 *3.15*	175 *6.89*	255 *10.039*	150 *5.91*	230 *9.055*
90.__.00750.088	87.5 *3.44*	182.5 *7.19*	270 *10.630*	157.5 *6.20*	245 *9.646*
• 90.__.00750.100	100 *3.94*	195 *7.68*	295 *11.614*	170 *6.69*	270 *10.630*
90.__.00750.113	112.5 *4.43*	207.5 *8.17*	320 *12.598*	182.5 *7.19*	295 *11.614*
• 90.__.00750.125	125 *4.92*	220 *8.66*	345 *13.583*	195 *7.68*	320 *12.598*
90.__.00750.138	137.5 *5.41*	232.5 *9.15*	370 *14.567*	207.5 *8.17*	345 *13.583*
90.__.00750.150	150 *5.91*	245 *9.65*	395 *15.551*	220 *8.66*	370 *14.567*
• 90.__.00750.160	160 *6.30*	255 *10.04*	415 *16.339*	230 *9.06*	390 *15.354*
90.__.00750.175	175 *6.89*	270 *10.63*	445 *17.520*	245 *9.65*	420 *16.535*
90.__.00750.200	200 *7.87*	295 *11.61*	495 *19.488*	270 *10.63*	470 *18.504*
90.__.00750.225	225 *8.86*	320 *12.60*	545 *21.457*	295 *11.61*	520 *20.472*
90.__.00750.250	250 *9.84*	345 *13.58*	595 *23.425*	320 *12.60*	570 *22.441*
90.__.00750.275	275 *10.83*	370 *14.57*	645 *25.394*	N/A	N/A
90.__.00750.300	300 *11.81*	395 *15.55*	695 *27.362*	N/A	N/A

• ISO standard stroke lengths (90.10.00750 only)

Force Charts

On-Contact Force

psi	lb.
2175	1655
2000	1522
1750	1331
1500	1141
1000	761
500	380
250	190



On-Contact Force

bar	daN
150	736
125	614
100	491
75	368
50	245
25	123
20	98

Ordering Example:

90.10.00750.025. TO. C. 150

Part Number: Includes Series (90.10 or 90.8), Model and Stroke Length.

Mount Option: TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

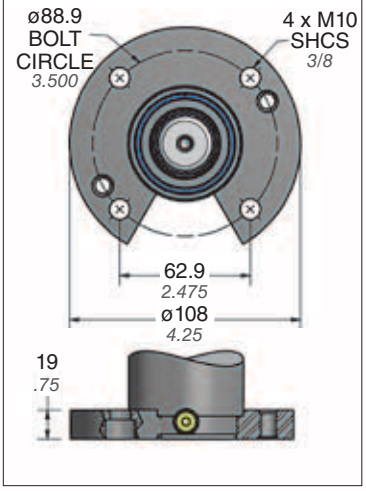
Charging Pressure: 15-150 bar *(220-2175 psi). When not specified, default is 150 bar.*

Operating System: C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

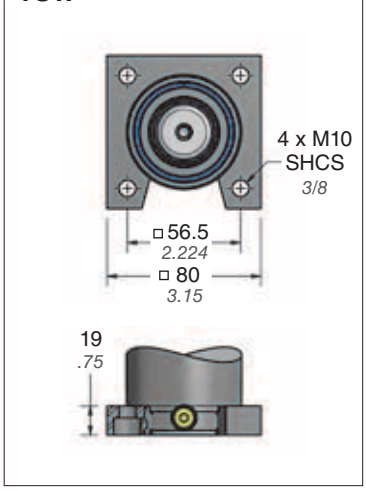
734.207.1100 • 800.DADCO.USA • fax 734.207.2222 • www.dadco.net

7.5 kN / 0.75 ton

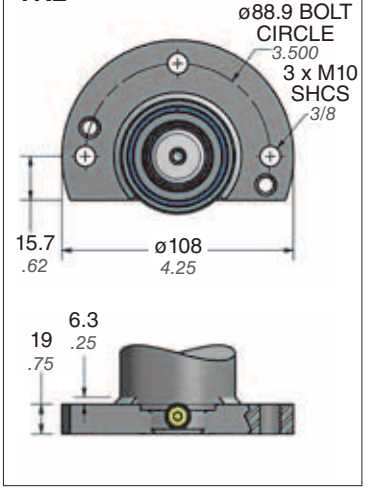
TFM — Welded



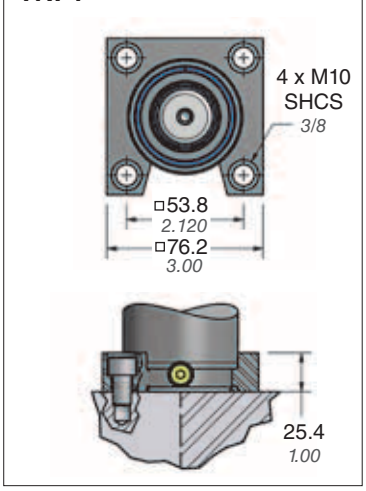
TSW — Welded



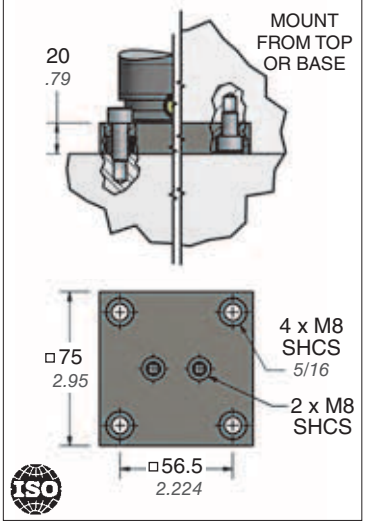
TR2 — Welded



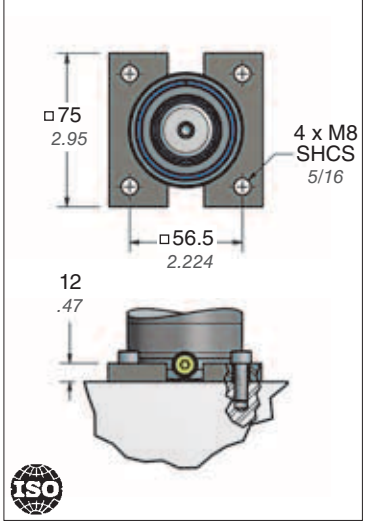
TNF1 — Welded



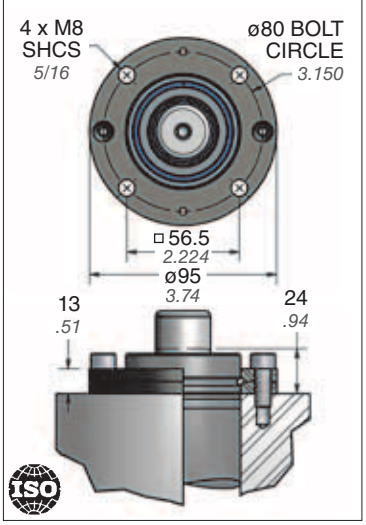
B11 — 90.11.00750



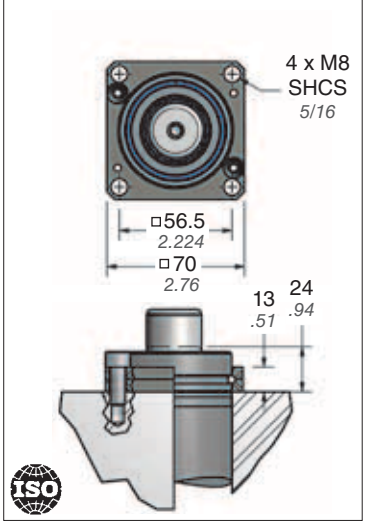
B12 — 90.12.00750



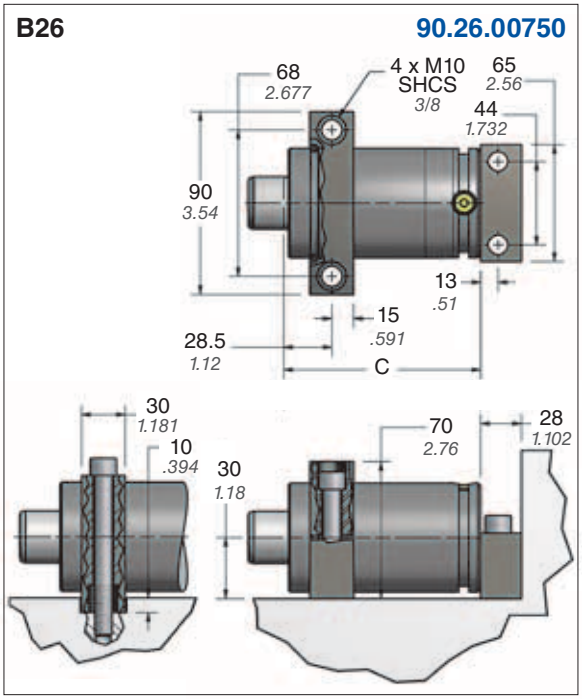
B21 — 90.21.00750



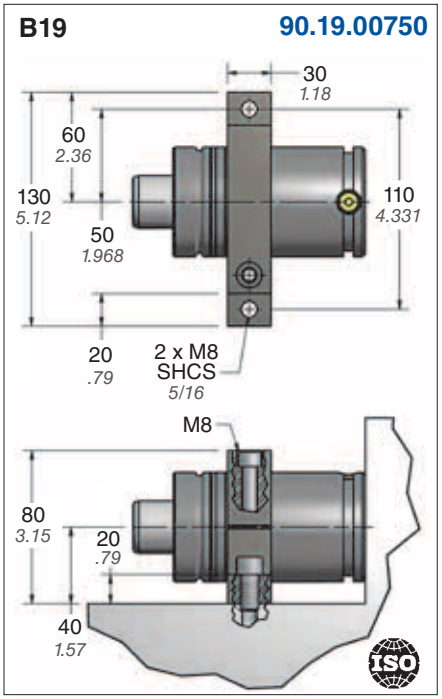
B25 — 90.25.00750



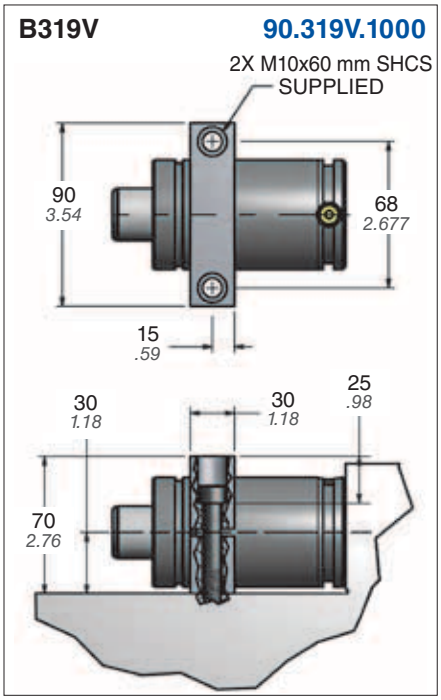
B26 — 90.26.00750



B19 — 90.19.00750

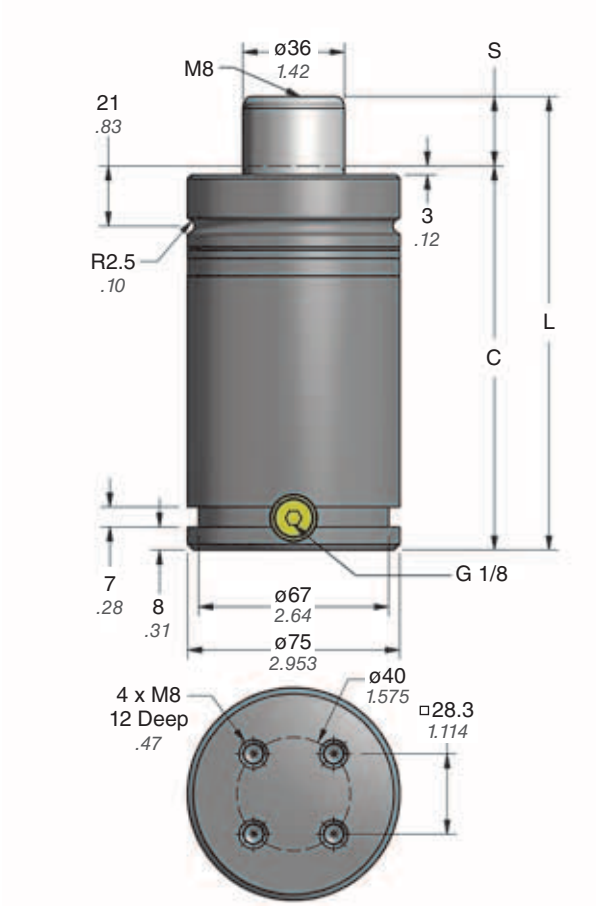


B319V — 90.319V.1000



Ordering Example: **Cylinder with Mount:** 90.10.00750.025.B21.C.150 **Mount Only:** 90.21.00750

15 kN / 1.5 ton



TO – Basic Model

25 mm shorter!

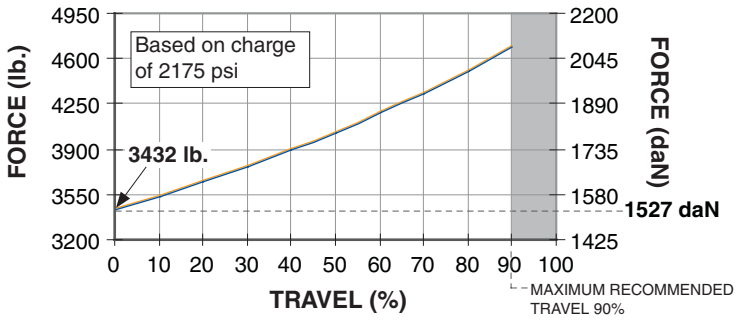
Part No.	S mm *inch*	90.10.01500 ISO C	90.10.01500 ISO L ±0.25 *±.010*	90.8.01500 C	90.8.01500 L ±0.25 *±.010*
90.__.01500.013	12.5 *0.49*	122.5 *4.82*	135 *5.315*	97.5 *3.84*	110 *4.331*
• 90.__.01500.025	25 *0.98*	135 *5.31*	160 *6.299*	110 *4.33*	135 *5.315*
90.__.01500.038	37.5 *1.48*	147.5 *5.81*	185 *7.283*	122.5 *4.82*	160 *6.299*
• 90.__.01500.050	50 *1.97*	160 *6.30*	210 *8.268*	135 *5.32*	185 *7.283*
90.__.01500.063	62.5 *2.46*	172.5 *6.79*	235 *9.252*	147.5 *5.81*	210 *8.268*
90.__.01500.075	75 *2.95*	185 *7.28*	260 *10.236*	160 *6.30*	235 *9.252*
• 90.__.01500.080	80 *3.15*	190 *7.48*	270 *10.630*	165 *6.50*	245 *9.646*
90.__.01500.088	87.5 *3.44*	197.5 *7.78*	285 *11.220*	172.5 *6.79*	260 *10.236*
• 90.__.01500.100	100 *3.94*	210 *8.27*	310 *12.205*	185 *7.28*	285 *11.220*
90.__.01500.113	112.5 *4.43*	222.5 *8.76*	335 *13.189*	197.5 *7.78*	310 *12.205*
• 90.__.01500.125	125 *4.92*	235 *9.25*	360 *14.173*	210 *8.27*	335 *13.189*
90.__.01500.138	137.5 *5.41*	247.5 *9.74*	385 *15.157*	222.5 *8.76*	360 *14.173*
90.__.01500.150	150 *5.91*	260 *10.24*	410 *16.142*	235 *9.25*	385 *15.157*
• 90.__.01500.160	160 *6.30*	270 *10.63*	430 *16.929*	245 *9.65*	405 *15.945*
90.__.01500.175	175 *6.89*	285 *11.22*	460 *18.110*	260 *10.24*	435 *17.126*
90.__.01500.200	200 *7.87*	310 *12.20*	510 *20.079*	285 *11.22*	485 *19.094*
90.__.01500.225	225 *8.86*	335 *13.19*	560 *22.047*	310 *12.21*	535 *21.063*
90.__.01500.250	250 *9.84*	360 *14.17*	610 *24.016*	335 *13.19*	585 *23.031*
90.__.01500.275	275 *10.83*	385 *15.16*	660 *25.984*	N/A	N/A
90.__.01500.300	300 *11.81*	410 *16.14*	710 *27.953*	N/A	N/A

• ISO standard stroke lengths (90.10.01500 only)

Force Charts

On-Contact Force

psi	lb.
2175	3432
2000	3155
1750	2761
1500	2367
1000	1578
500	789
250	394



On-Contact Force

bar	daN
150	1527
125	1272
100	1018
75	763
50	509
25	254
20	204

Ordering Example:

90.10.01500.025. *TO*. *C*. *150*

Part Number: Includes Series (90.10 or 90.8), Model and Stroke Length.

Mount Option: TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

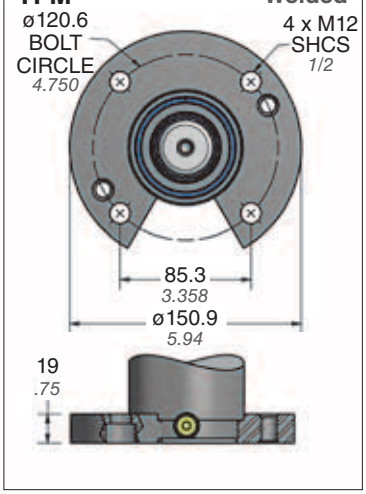
Charging Pressure: 15-150 bar *(220-2175 psi). When not specified, default is 150 bar.*

Operating System: C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

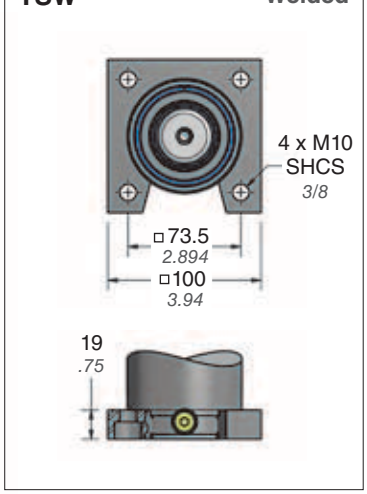
734.207.1100 • 800.DADCO.USA • fax 734.207.2222 • www.dadco.net

15 kN / 1.5 ton

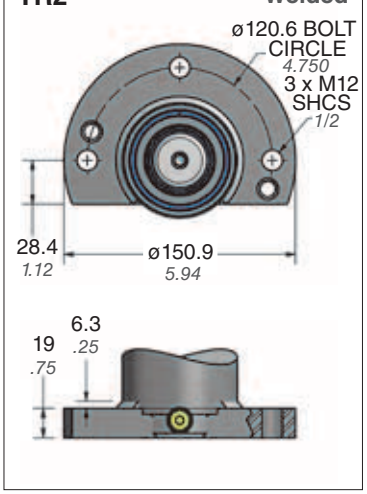
TFM — Welded



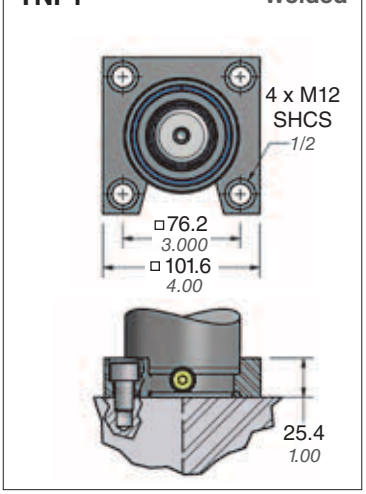
TSW — Welded



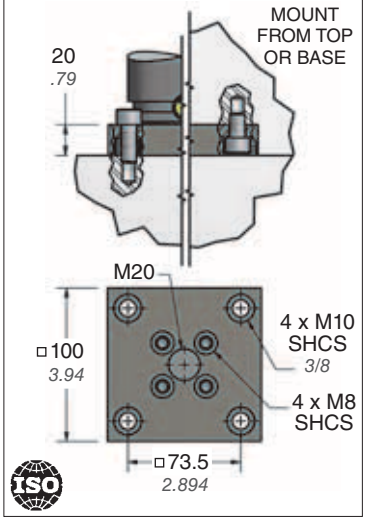
TR2 — Welded



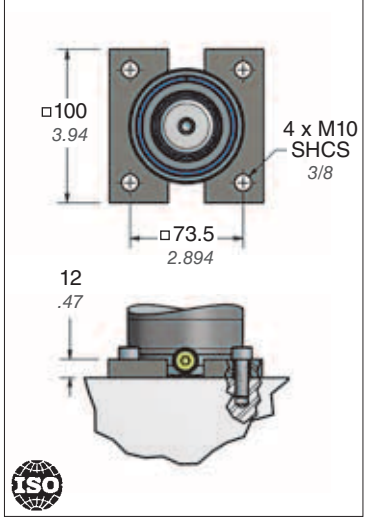
TNF1 — Welded



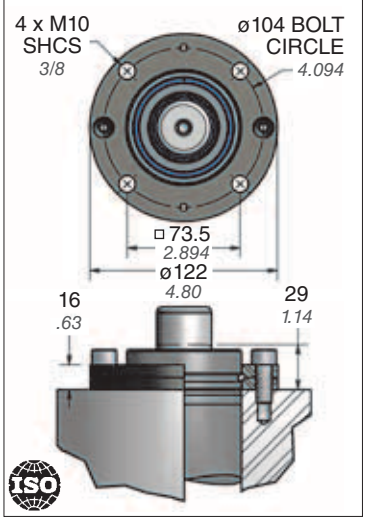
B11 — 90.11.01500



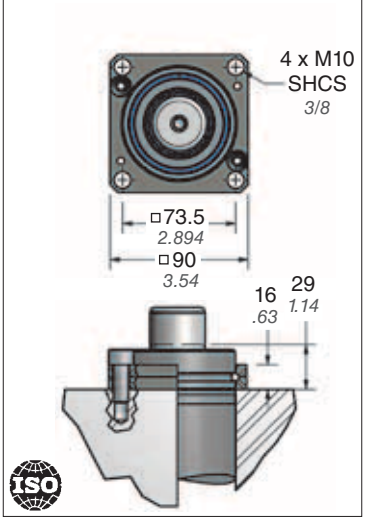
B12 — 90.12.01500



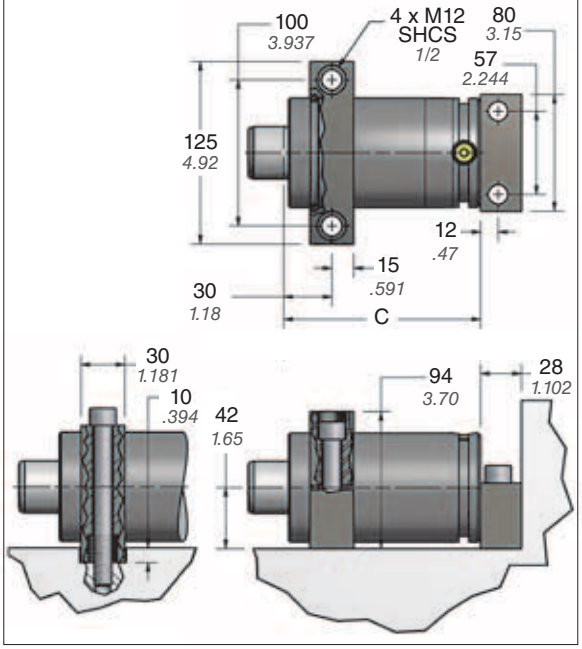
B21 — 90.21.01500



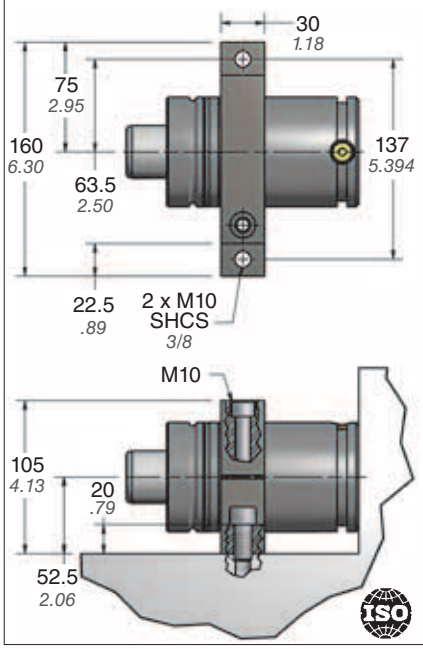
B25 — 90.25.01500



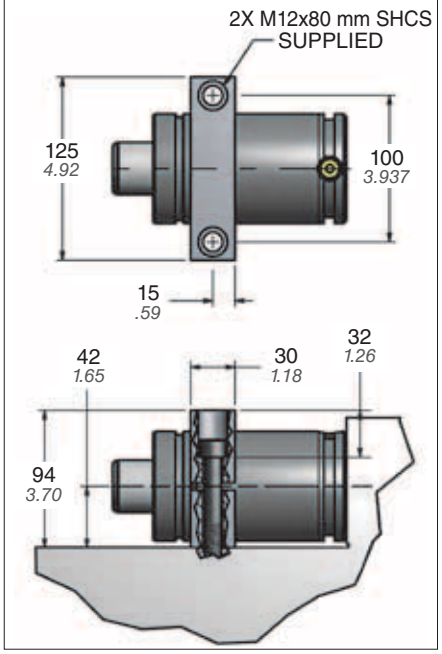
B26 — 90.26.01500



B19 — 90.19.01500



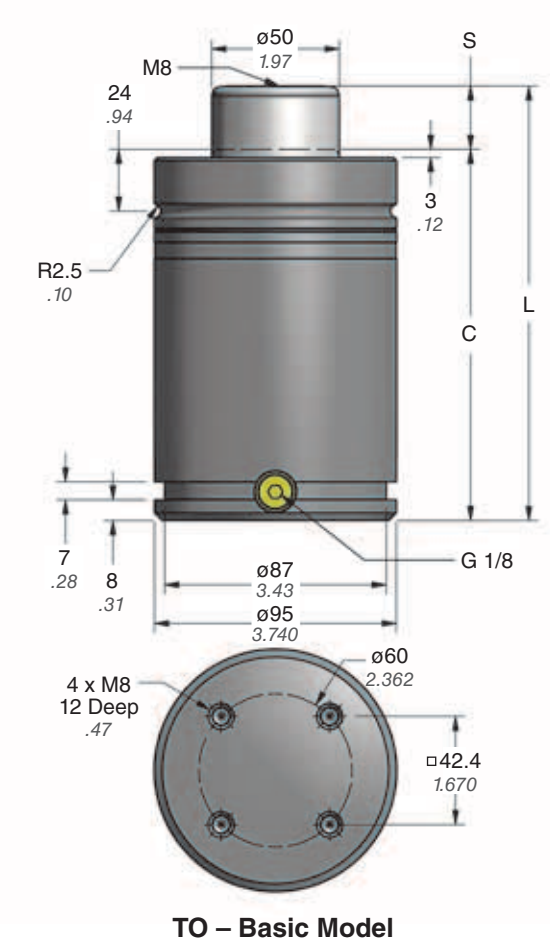
B319V — 90.319V.2600



Ordering Example: **Cylinder with Mount:** 90.10.01500.025.B21.C.150 **Mount Only:** 90.21.01500

90.10 and 90.8 Series Nitrogen Gas Springs

30 kN / 3 ton



TO – Basic Model

25 mm shorter!

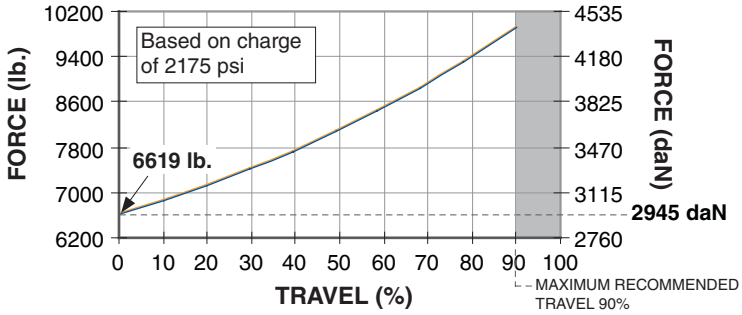
Part No.	S mm *inch*	90.10.03000 ISO C	90.10.03000 ISO L ±0.25 *±.010*	90.8.03000 C	90.8.03000 L ±0.25 *±.010*
90.__.03000.013	12.5 *0.49*	132.5 *5.22*	145 *5.709*	107.5 *4.23*	120 *4.724*
• 90.__.03000.025	25 *0.98*	145 *5.71*	170 *6.693*	120 *4.72*	145 *5.709*
90.__.03000.038	37.5 *1.48*	157.5 *6.20*	195 *7.677*	132.5 *5.22*	170 *6.693*
• 90.__.03000.050	50 *1.97*	170 *6.69*	220 *8.661*	145 *5.71*	195 *7.677*
90.__.03000.063	62.5 *2.46*	182.5 *7.19*	245 *9.646*	157.5 *6.20*	220 *8.661*
90.__.03000.075	75 *2.95*	195 *7.68*	270 *10.630*	170 *6.69*	245 *9.646*
• 90.__.03000.080	80 *3.15*	200 *7.87*	280 *11.024*	175 *6.89*	255 *10.039*
90.__.03000.088	87.5 *3.44*	207.5 *8.17*	295 *11.614*	182.5 *7.19*	270 *10.630*
• 90.__.03000.100	100 *3.94*	220 *8.66*	320 *12.598*	195 *7.68*	295 *11.614*
90.__.03000.113	112.5 *4.43*	232.5 *9.15*	345 *13.583*	207.5 *8.17*	320 *12.598*
• 90.__.03000.125	125 *4.92*	245 *9.65*	370 *14.567*	220 *8.66*	345 *13.583*
90.__.03000.138	137.5 *5.41*	257.5 *10.14*	395 *15.551*	232.5 *9.15*	370 *14.567*
90.__.03000.150	150 *5.91*	270 *10.63*	420 *16.535*	245 *9.65*	395 *15.551*
• 90.__.03000.160	160 *6.30*	280 *11.02*	440 *17.323*	255 *10.04*	415 *16.339*
90.__.03000.175	175 *6.89*	295 *11.61*	470 *18.504*	270 *10.63*	445 *17.520*
90.__.03000.200	200 *7.87*	320 *12.60*	520 *20.472*	295 *11.61*	495 *19.488*
90.__.03000.225	225 *8.86*	345 *13.58*	570 *22.441*	320 *12.60*	545 *21.457*
90.__.03000.250	250 *9.84*	370 *14.57*	620 *24.409*	345 *13.58*	595 *23.425*
90.__.03000.275	275 *10.83*	395 *15.55*	670 *26.378*	N/A	N/A
90.__.03000.300	300 *11.81*	420 *16.54*	720 *28.346*	N/A	N/A

• ISO standard stroke lengths (90.10.03000 only)

Force Charts

On-Contact Force

psi	lb.
2175	6619
2000	6087
1750	5326
1500	4565
1000	3043
500	1522
250	761



On-Contact Force

bar	daN
150	2945
125	2454
100	1963
75	1473
50	982
25	491
20	393

Ordering Example:

90.10.03000.025. *TO*. *C*. *150*

Part Number: Includes Series (90.10 or 90.8), Model and Stroke Length.

Mount Option: TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

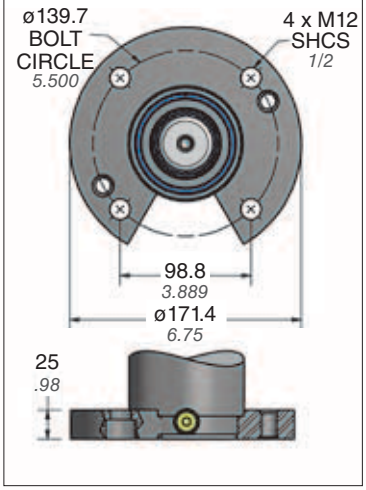
Charging Pressure: 15-150 bar *(220-2175 psi). When not specified, default is 150 bar.*

Operating System: C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

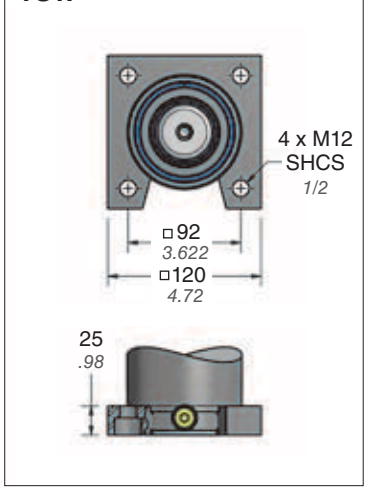
734.207.1100 • 800.DADCO.USA • fax 734.207.2222 • www.dadco.net

30 kN / 3 ton

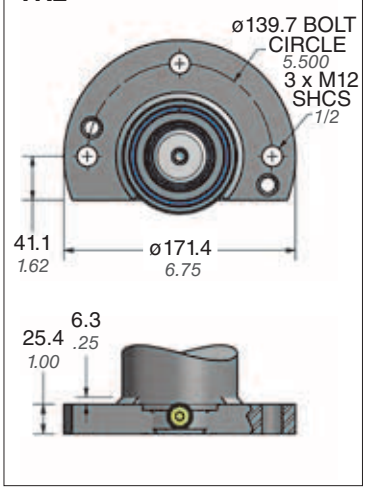
TFM — Welded



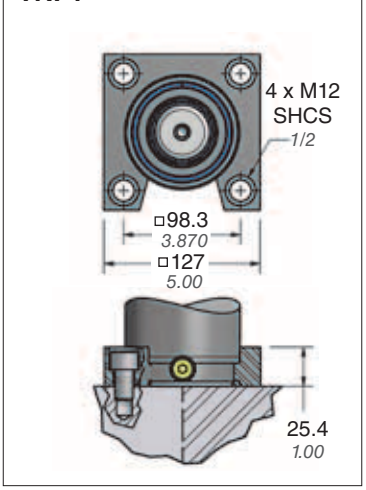
TSW — Welded



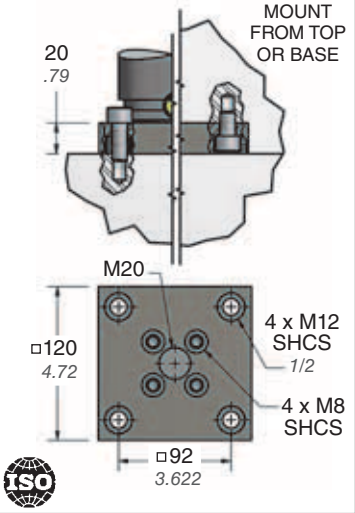
TR2 — Welded



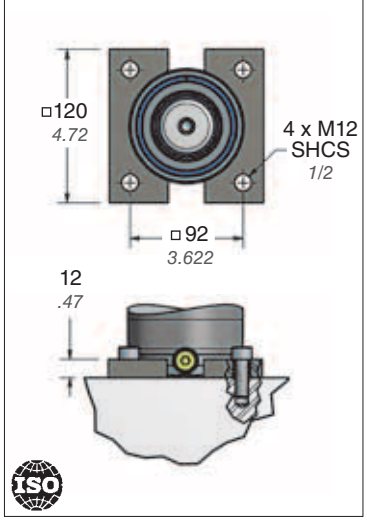
TNF1 — Welded



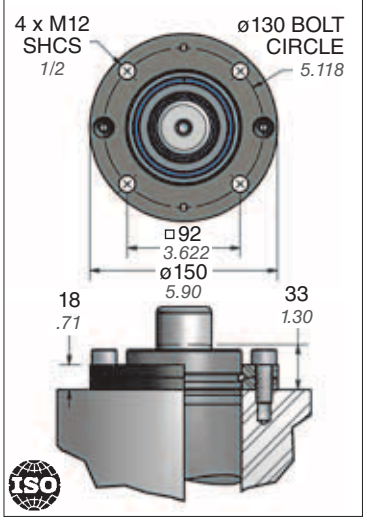
B11 — 90.11.03000



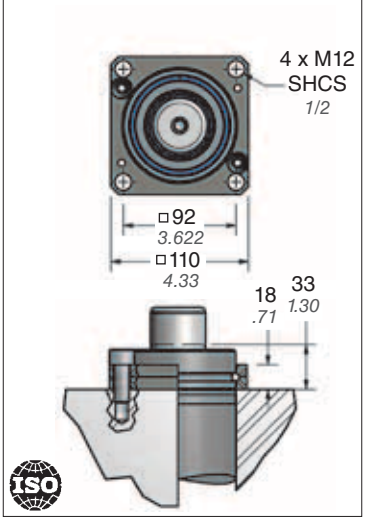
B12 — 90.12.03000



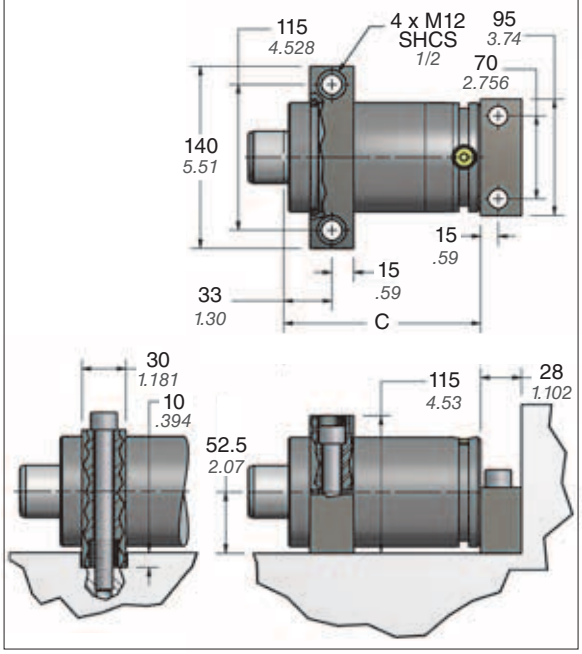
B21 — 90.21.03000



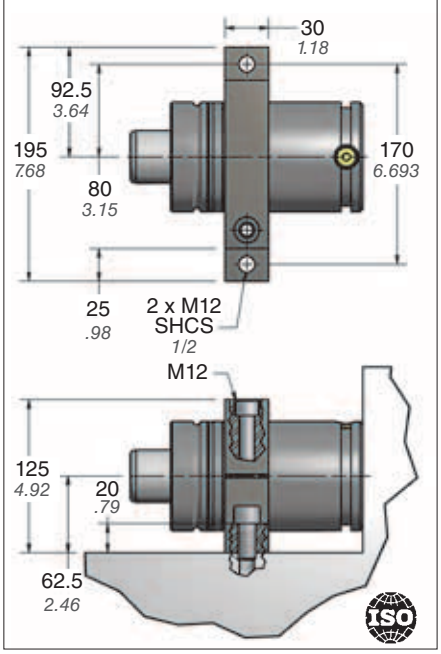
B25 — 90.25.03000



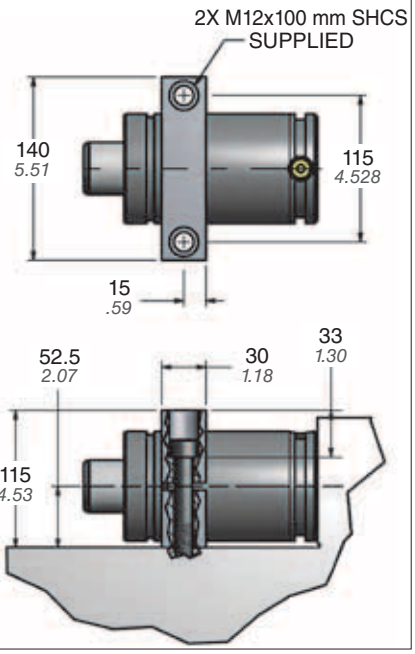
B26 — 90.26.03000



B19 — 90.19.03000

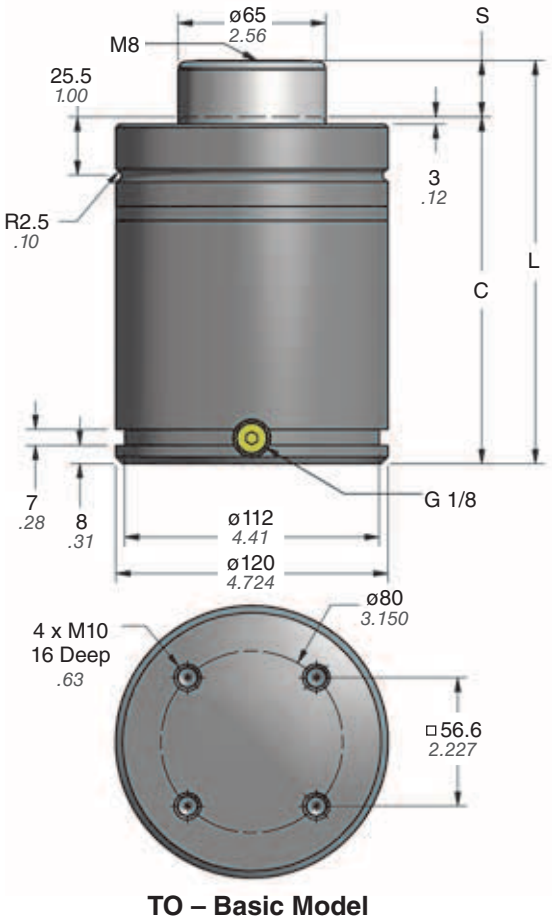


B319V — 90.319V.4600



Ordering Example: **Cylinder with Mount:** 90.10.03000.025.B21.C.150 **Mount Only:** 90.21.03000

50 kN / 5 ton



TO – Basic Model

37.5 mm shorter!

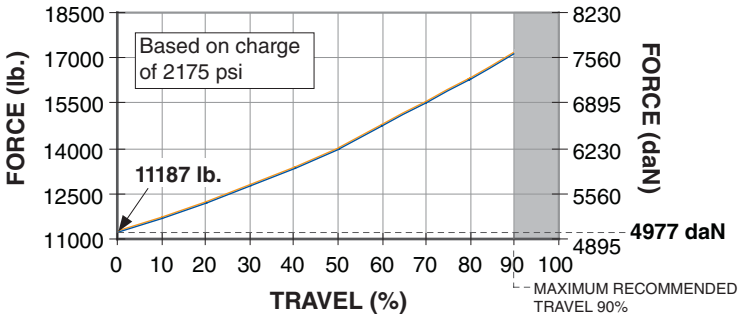
Part No.	S mm *inch*	90.10.05000 ISO C	90.10.05000 ISO L ±0.25 *±.010*	90.8.05000 C	90.8.05000 L ±0.25 *±.010*
• 90.__.05000.025	25 *0.98*	165 *6.50*	190 *7.480*	127.5 *5.02*	152.5 *6.004*
90.__.05000.038	37.5 *1.48*	177.5 *6.99*	215 *8.465*	140 *5.51*	177.5 *6.988*
• 90.__.05000.050	50 *1.97*	190 *7.48*	240 *9.449*	152.5 *6.00*	202.5 *7.972*
90.__.05000.063	62.5 *2.46*	202.5 *7.97*	265 *10.433*	165 *6.50*	227.5 *8.957*
90.__.05000.075	75 *2.95*	215 *8.46*	290 *11.417*	177.5 *6.99*	252.5 *9.941*
• 90.__.05000.080	80 *3.15*	220 *8.66*	300 *11.811*	182.5 *7.19*	262.5 *10.335*
90.__.05000.088	87.5 *3.44*	227.5 *8.96*	315 *12.402*	190 *7.48*	277.5 *10.925*
• 90.__.05000.100	100 *3.94*	240 *9.45*	340 *13.386*	202.5 *7.97*	302.5 *11.909*
90.__.05000.113	112.5 *4.43*	252.5 *9.94*	365 *14.370*	215 *8.47*	327.5 *12.894*
• 90.__.05000.125	125 *4.92*	265 *10.43*	390 *15.354*	227.5 *8.96*	352.5 *13.878*
90.__.05000.138	137.5 *5.41*	277.5 *10.93*	415 *16.339*	240 *9.45*	377.5 *14.862*
90.__.05000.150	150 *5.91*	290 *11.42*	440 *17.323*	252.5 *9.94*	402.5 *15.846*
• 90.__.05000.160	160 *6.30*	300 *11.81*	460 *18.110*	262.5 *10.34*	422.5 *16.634*
90.__.05000.175	175 *6.89*	315 *12.40*	490 *19.291*	277.5 *10.93*	452.5 *17.815*
90.__.05000.200	200 *7.87*	340 *13.39*	540 *21.260*	302.5 *11.91*	502.5 *19.783*
90.__.05000.225	225 *8.86*	365 *14.37*	590 *23.228*	327.5 *12.89*	552.5 *21.752*
90.__.05000.250	250 *9.84*	390 *15.35*	640 *25.197*	352.5 *13.88*	602.5 *23.720*
90.__.05000.275	275 *10.83*	415 *16.34*	690 *27.165*	N/A	N/A
90.__.05000.300	300 *11.81*	440 *17.32*	740 *29.134*	N/A	N/A

• ISO standard stroke lengths (90.10.05000 only)

Force Charts

On-Contact Force

psi	lb.
2175	11187
2000	10287
1750	9001
1500	7715
1000	5143
500	2572
250	1286



On-Contact Force

bar	daN
150	4977
125	4148
100	3318
75	2489
50	1659
25	830
20	664

Ordering Example:

90.10.05000.025. *TO*. *C*. *150*

Part Number: Includes Series (90.10 or 90.8), Model and Stroke Length.

Mount Option: TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

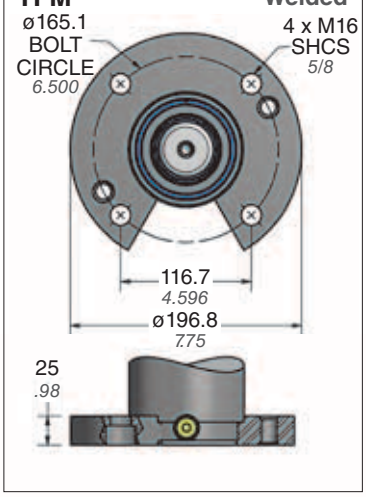
Charging Pressure: 15-150 bar *(220-2175 psi). When not specified, default is 150 bar.*

Operating System: C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

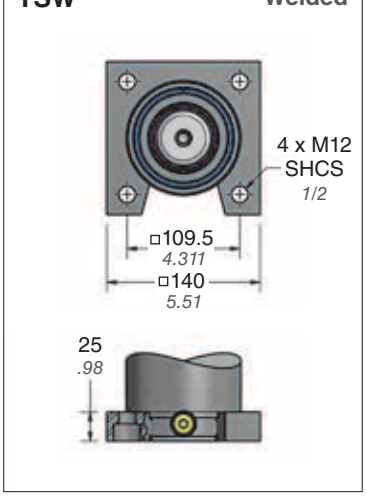
734.207.1100 • 800.DADCO.USA • fax 734.207.2222 • www.dadco.net

50 kN / 5 ton

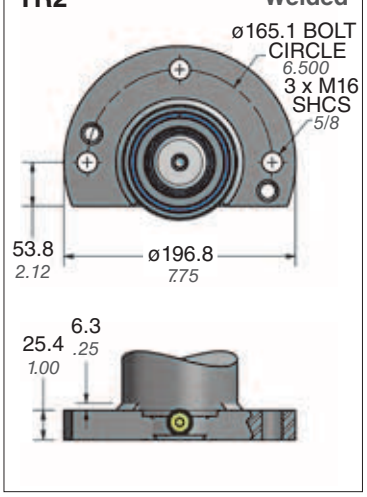
TFM — Welded



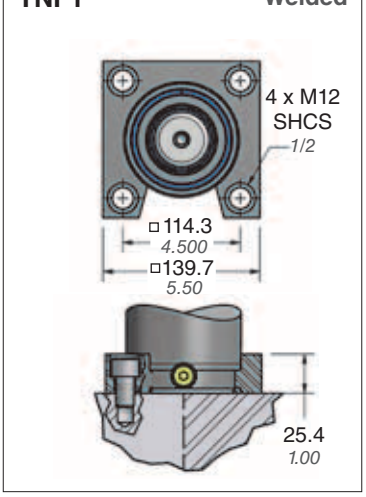
TSW — Welded



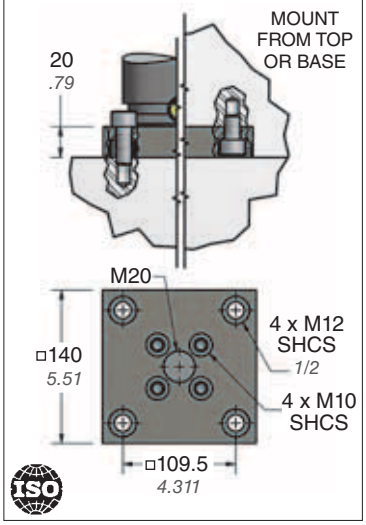
TR2 — Welded



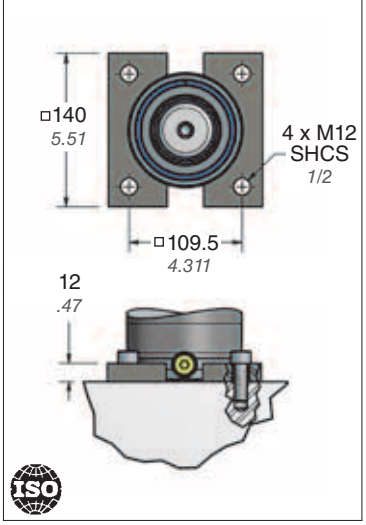
TNF1 — Welded



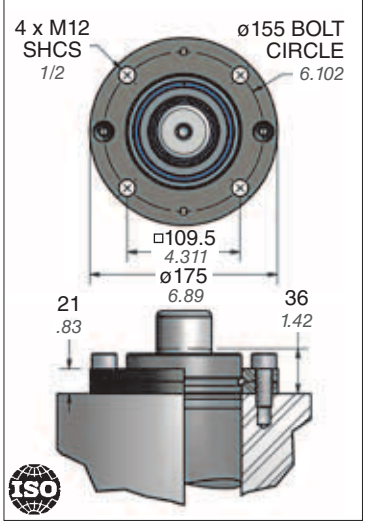
B11 — 90.11.05000



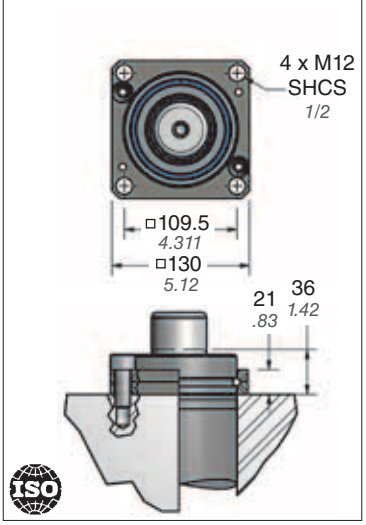
B12 — 90.12.05000



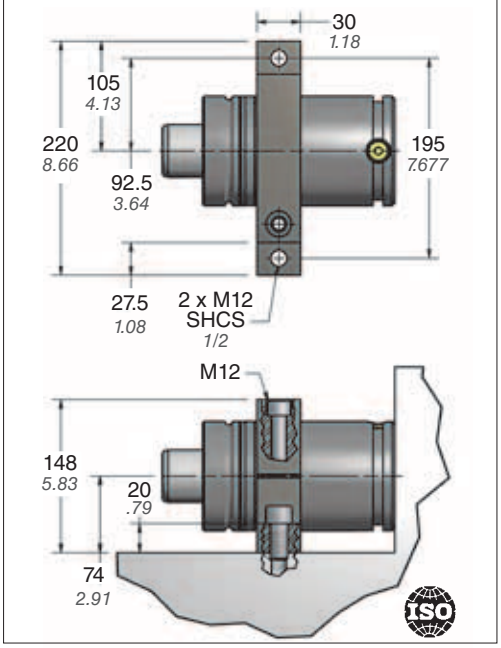
B21 — 90.21.05000



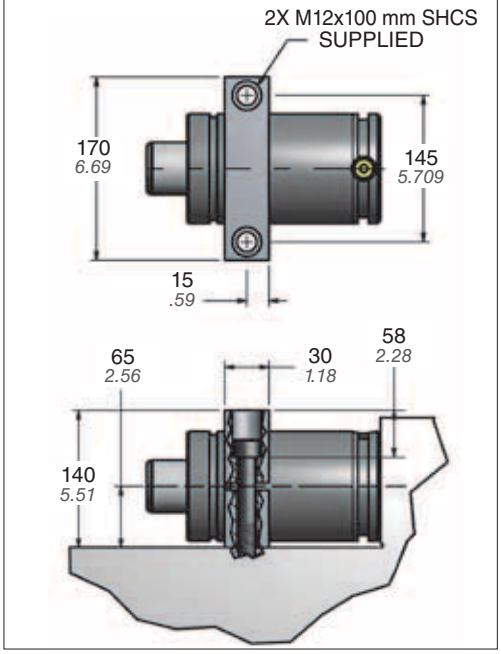
B25 — 90.25.05000



B19 — 90.19.05000



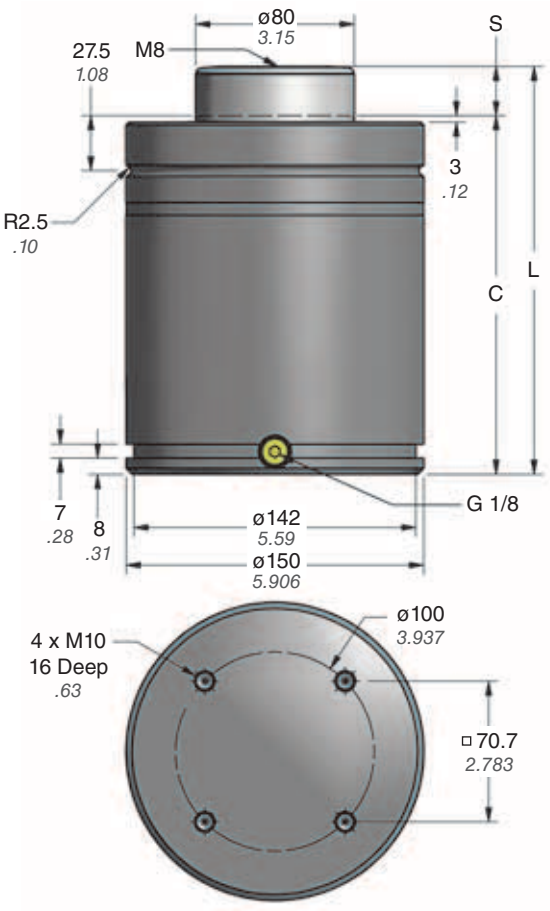
B319V — 90.319V.6600



Ordering Example: **Cylinder with Mount:** 90.10.05000.025.B21.C.150 **Mount Only:** 90.21.05000

90.10 and 90.8 Series Nitrogen Gas Springs

75 kN / 7.5 ton



TO – Basic Model

50 mm shorter!

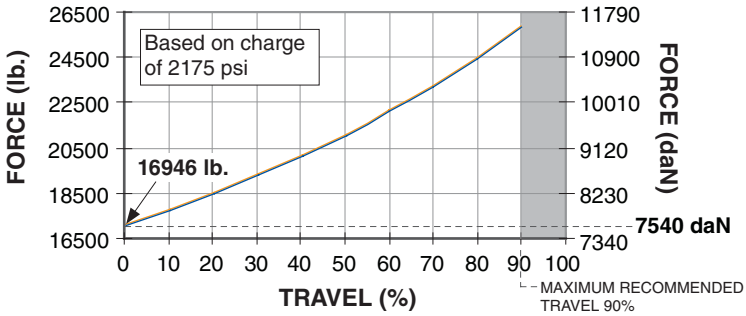
Part No.	S mm *inch*	90.10.07500 ISO C	90.10.07500 ISO L ±0.25 *±.010*	90.8.07500 C	90.8.07500 L ±0.25 *±.010*
• 90.__.07500.025	25 *0.98*	180 *7.09*	205 *8.071*	130 *5.12*	155 *6.102*
90.__.07500.038	37.5 *1.48*	192.5 *7.58*	230 *9.055*	142.5 *5.61*	180 *7.087*
• 90.__.07500.050	50 *1.97*	205 *8.07*	255 *10.039*	155 *6.10*	205 *8.071*
90.__.07500.063	62.5 *2.46*	217.5 *8.56*	280 *11.024*	167.5 *6.59*	230 *9.055*
90.__.07500.075	75 *2.95*	230 *9.06*	305 *12.008*	180 *7.09*	255 *10.039*
• 90.__.07500.080	80 *3.15*	235 *9.25*	315 *12.402*	185 *7.28*	265 *10.433*
90.__.07500.088	87.5 *3.44*	242.5 *9.55*	330 *12.992*	192.5 *7.58*	280 *11.024*
• 90.__.07500.100	100 *3.94*	255 *10.04*	355 *13.976*	205 *8.07*	305 *12.008*
90.__.07500.113	112.5 *4.43*	267.5 *10.53*	380 *14.961*	217.5 *8.56*	330 *12.992*
• 90.__.07500.125	125 *4.92*	280 *11.02*	405 *15.945*	230 *9.06*	355 *13.976*
90.__.07500.138	137.5 *5.41*	292.5 *11.52*	430 *16.929*	242.5 *9.55*	380 *14.961*
90.__.07500.150	150 *5.91*	305 *12.01*	455 *17.913*	255 *10.04*	405 *15.945*
• 90.__.07500.160	160 *6.30*	315 *12.40*	475 *18.701*	265 *10.43*	425 *16.732*
90.__.07500.175	175 *6.89*	330 *12.99*	505 *19.882*	280 *11.02*	455 *17.913*
90.__.07500.200	200 *7.87*	355 *13.98*	555 *21.850*	305 *12.01*	505 *19.882*
90.__.07500.225	225 *8.86*	380 *14.96*	605 *23.819*	330 *12.99*	555 *21.850*
90.__.07500.250	250 *9.84*	405 *15.94*	655 *25.787*	355 *13.98*	605 *23.819*
90.__.07500.275	275 *10.83*	430 *16.93*	705 *27.756*	N/A	N/A
90.__.07500.300	300 *11.81*	455 *17.91*	755 *29.724*	N/A	N/A

• ISO standard stroke lengths (90.10.07500 only)

Force Charts

On-Contact Force

psi	lb.
2175	16946
2000	15582
1750	13635
1500	11687
1000	7791
500	3896
250	1948



On-Contact Force

bar	daN
150	7540
125	6283
100	5027
75	3770
50	2513
25	1257
20	1005

Ordering Example:

90.10.07500.025. *TO*. *C*. *150*

Part Number: Includes Series (90.10 or 90.8), Model and Stroke Length.

Mount Option: TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

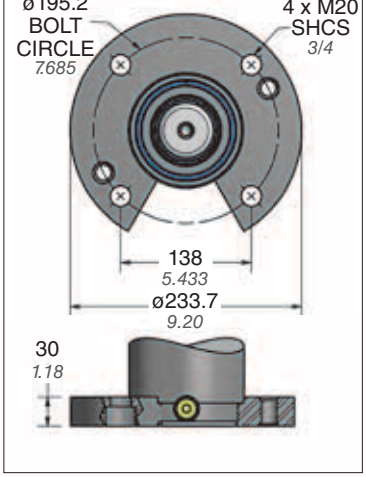
Charging Pressure: 15-150 bar *(220-2175 psi). When not specified, default is 150 bar.*

Operating System: C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

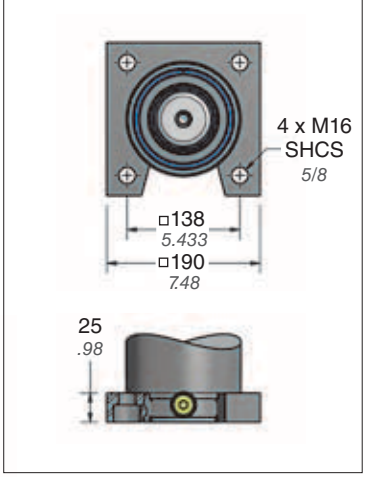
734.207.1100 • 800.DADCO.USA • fax 734.207.2222 • www.dadco.net

75 kN / 7.5 ton

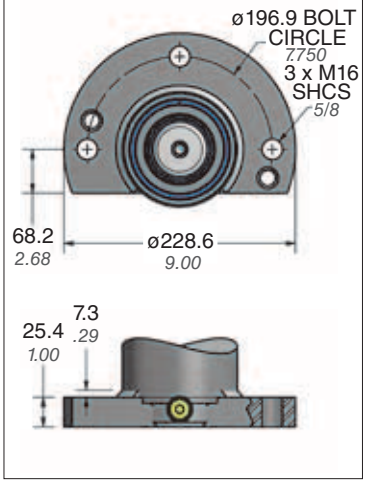
TFM — Welded



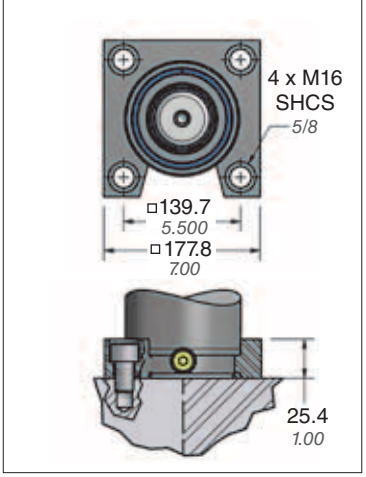
TSW — Welded



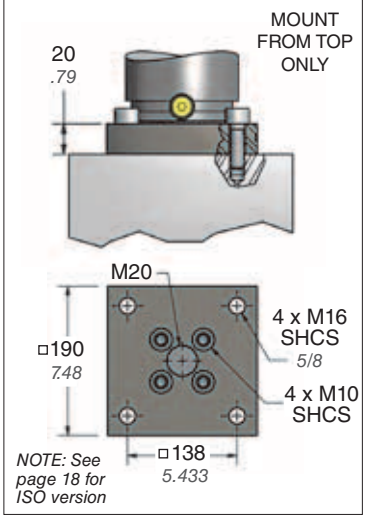
TR2 — Welded



TNF1 — Welded

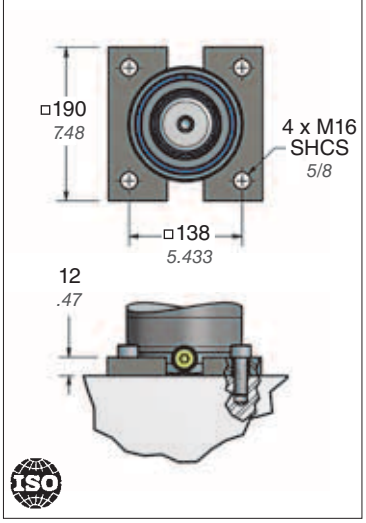


B11 — 90.11.07500

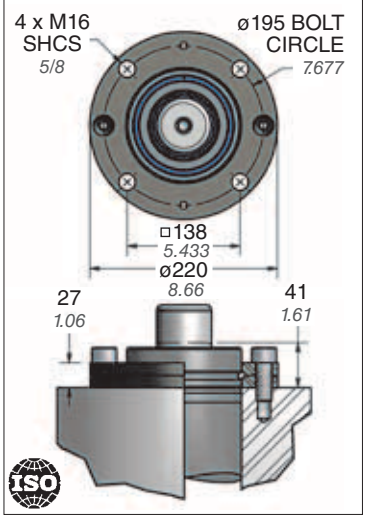


NOTE: See page 18 for ISO version

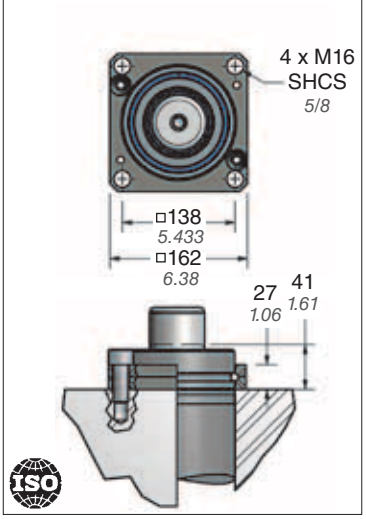
B12 — 90.12.07500



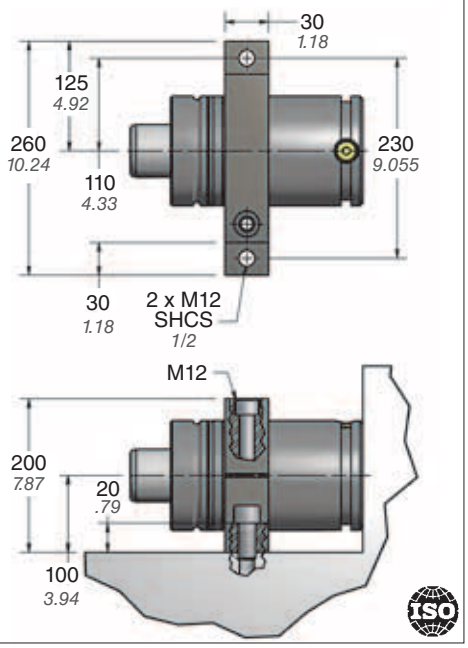
B21 — 90.21.07500



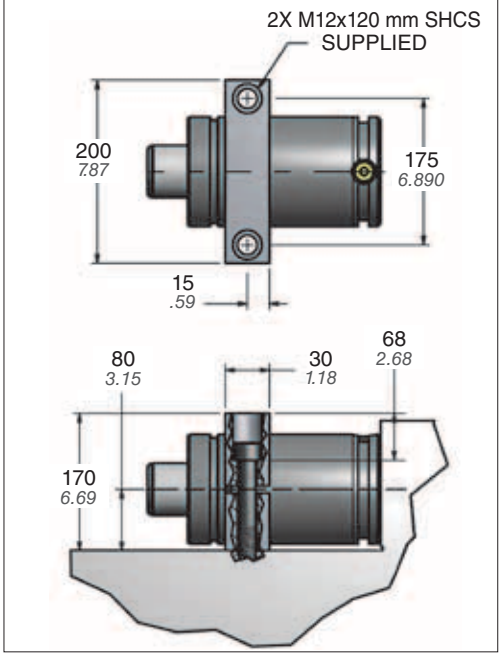
B25 — 90.25.07500



B19 — 90.19.07500

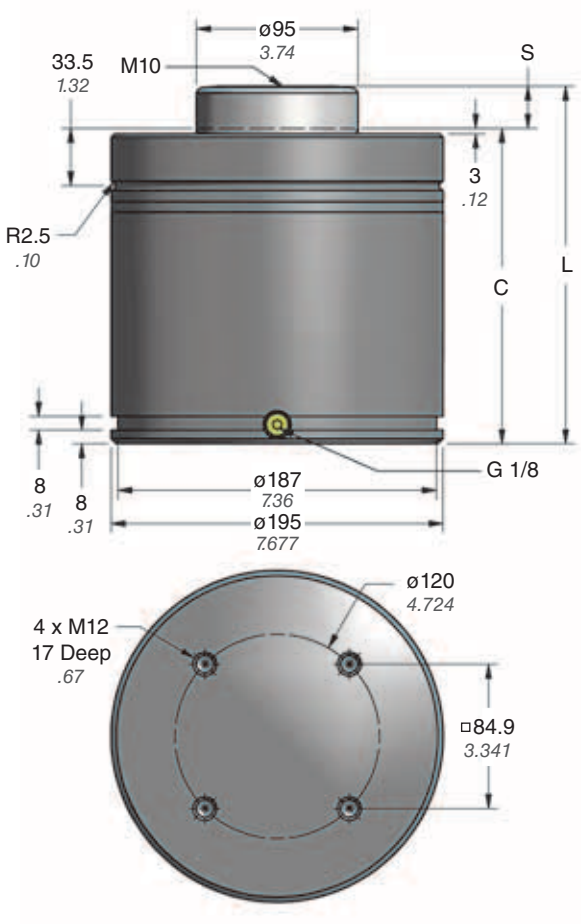


B319V — 90.319V.9600



Ordering Example: **Cylinder with Mount:** 90.10.07500.025.B21.C.150 **Mount Only:** 90.21.07500

100 kN / 10 ton



TO – Basic Model

Model Not Available in 90.8 Series

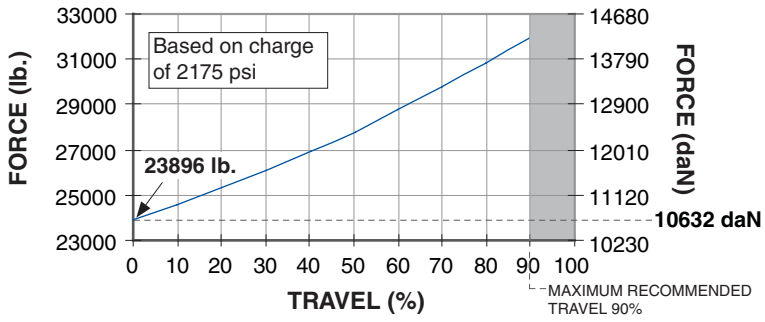
90.10.10000 ISO			
Part No.	**S** mm *inch*	**C**	**L** ±0.25 *±.010*
90.10.10000.025	25 *0.98*	185 *7.28*	210 *8.268*
90.10.10000.038	37.5 *1.48*	197.5 *7.78*	235 *9.252*
• 90.10.10000.050	50 *1.97*	210 *8.27*	260 *10.236*
90.10.10000.063	62.5 *2.46*	222.5 *8.76*	285 *11.220*
• 90.10.10000.080	80 *3.15*	240 *9.45*	320 *12.598*
• 90.10.10000.100	100 *3.94*	260 *10.24*	360 *14.173*
• 90.10.10000.125	125 *4.92*	285 *11.22*	410 *16.142*
• 90.10.10000.160	160 *6.30*	320 *12.60*	480 *18.898*
• 90.10.10000.200	200 *7.87*	360 *14.17*	560 *22.047*
• 90.10.10000.250	250 *9.84*	410 *16.14*	660 *25.984*

• ISO standard stroke lengths

Force Charts

On-Contact Force

psi	lb.
2175	23896
2000	21974
1750	19227
1500	16480
1000	10987
500	5493
250	2747



On-Contact Force

bar	daN
150	10632
125	8860
100	7088
75	5316
50	3544
25	1772
20	1418

Ordering Example:

90.10.10000.025. *TO*. *C*. *150*

Part Number: Includes Series (90.10), Model and Stroke Length.

Mount Option: TO = Basic Model. *When not specified, default is TO.* Mounts ordered with cylinder will be attached at the factory.

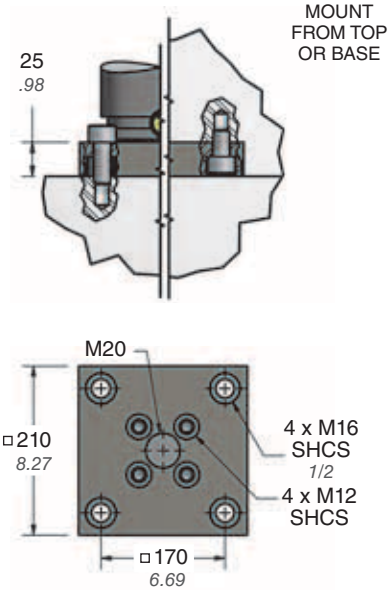
Charging Pressure: 15-150 bar *(220-2175 psi). When not specified, default is 150 bar.*

Operating System: C = Self-contained, F = Open Flow Fitting. *When not specified, default is C, self-contained.*

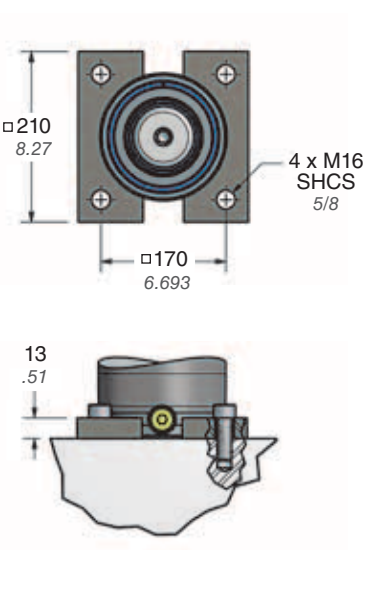
734.207.1100 • 800.DADCO.USA • fax 734.207.2222 • www.dadco.net

100 kN / 10 ton

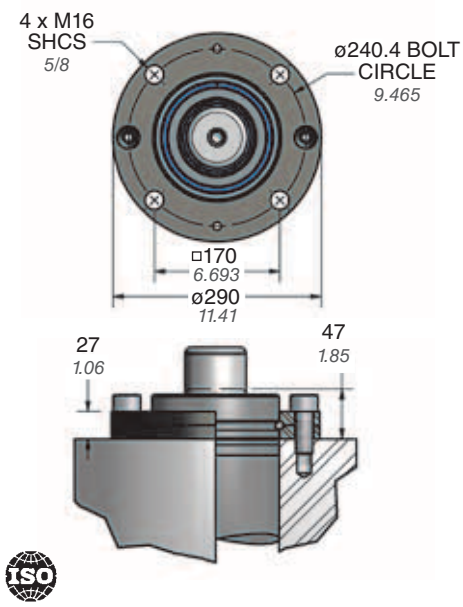
B11 — 90.11.10000



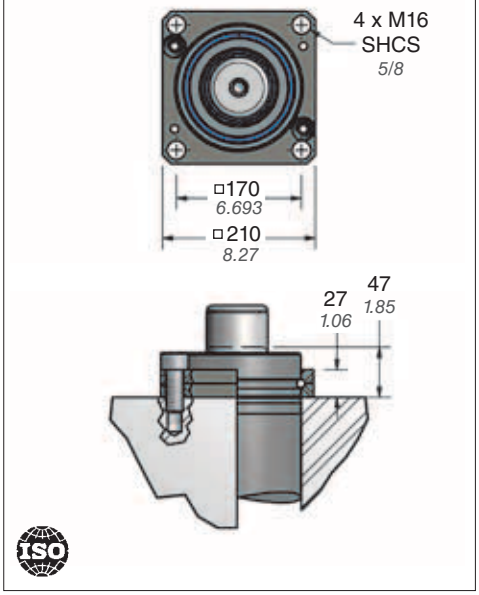
B12 — 90.12.10000



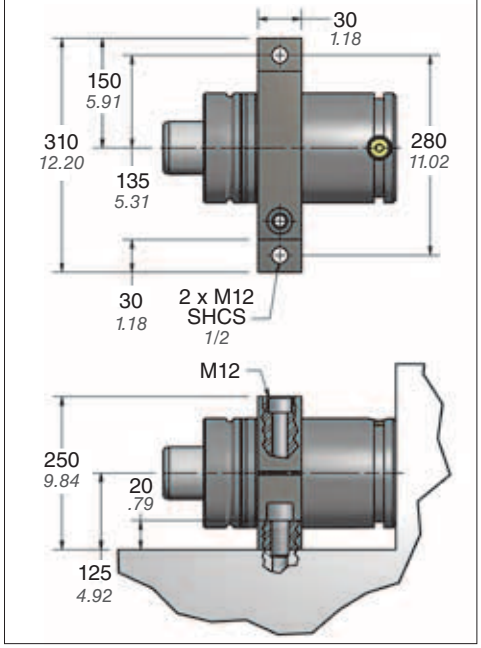
B21 — 90.21.10000



B25 — 90.25.10000



B19 — 90.19.10000

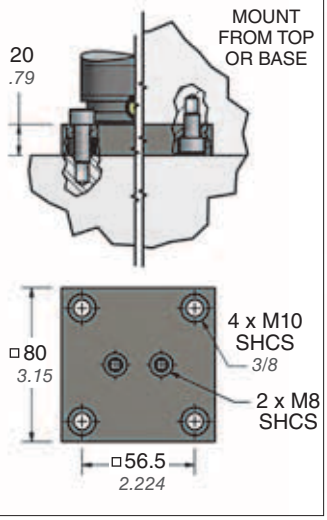


Ordering Example: **Cylinder with Mount:** 90.10.10000.050.B21.C.150 **Mount Only:** 90.21.10000

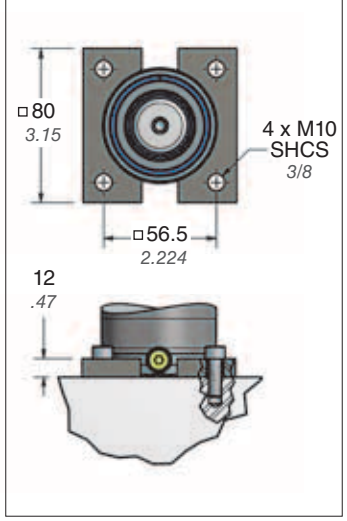
90.10 Miscellaneous Mounts

Special WDX Mounts

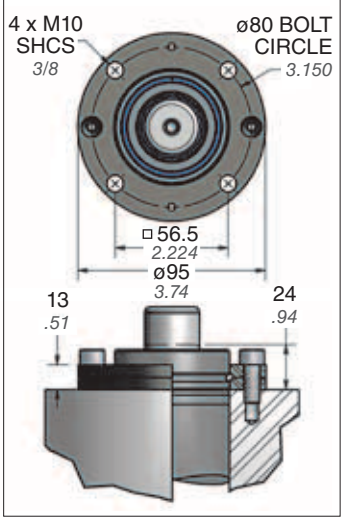
F11 **90.911.00750**



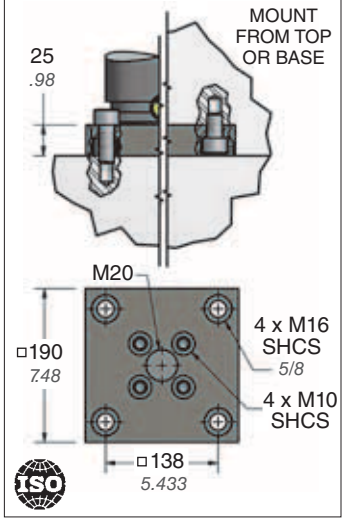
F12 **90.912.00750**



F21 **90.921.00750**

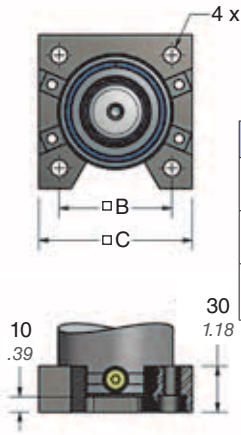


F11 **90.911.07500**



Special WDX Mounts (Continued)

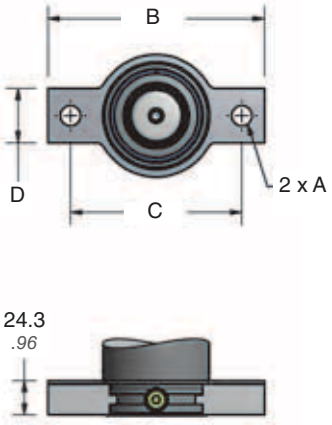
B31 **90.31.____**



	00750	01500	03000	05000	07500
A	M10 3/8	M10 3/8	M12 1/2	M12 1/2	M16 5/8
B	56.5 2.224	73.5 2.894	92 3.622	109.5 4.311	138 5.433
C	80 3.15	100 3.94	120 4.72	140 5.51	190 7.48

Alternative Mounts

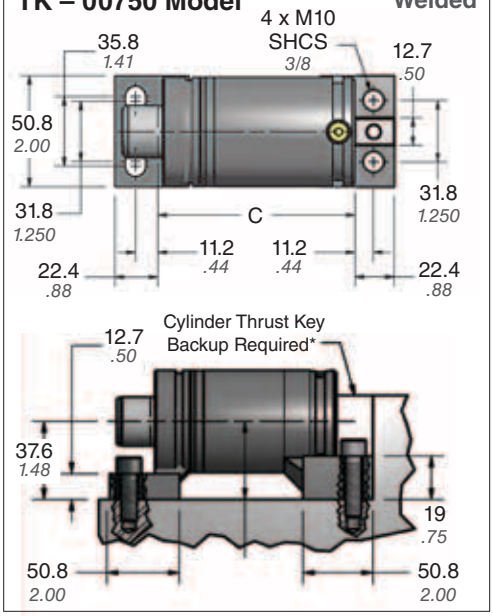
B14 **90.14.____**



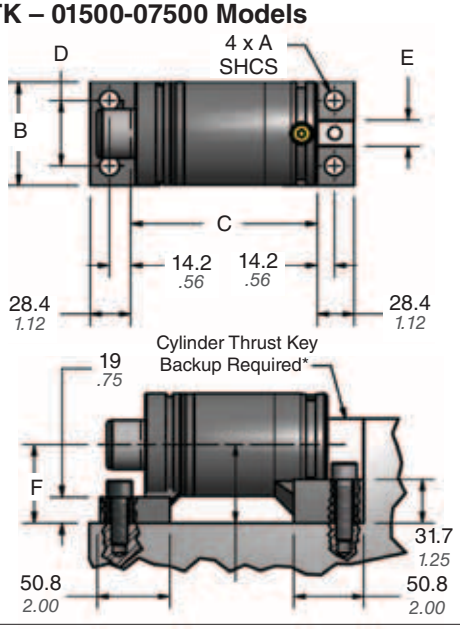
	00750	01500	03000	05000
A	M10 3/8	M12 1/2	M12 1/2	M12 1/2
B	108 4.25	152.4 6.00	177.8 7.00	196.8 7.75
C	88.9 3.500	120.6 4.750	146 5.750	165.1 6.500
D	31.7 1.25	38.1 1.50	38.1 1.50	38.1 1.50

Alternative Mounts (Continued)

TK – 00750 Model Welded



TK – 01500-07500 Models Welded



	01500	03000	05000	07500
A	M12 1/2	M12 1/2	M16 5/8	M16 5/8
B	76.2 3.00	95.2 3.75	120.6 4.75	152.4 6.00
D	47.6 1.875	63.5 2.500	88.9 3.500	120.7 4.750
E	19 .75	25.4 1.00	31.7 1.25	38.1 1.50
F	56.5 2.22	66.5 2.62	79 3.11	94 3.70

** Thrust Key not supplied.*

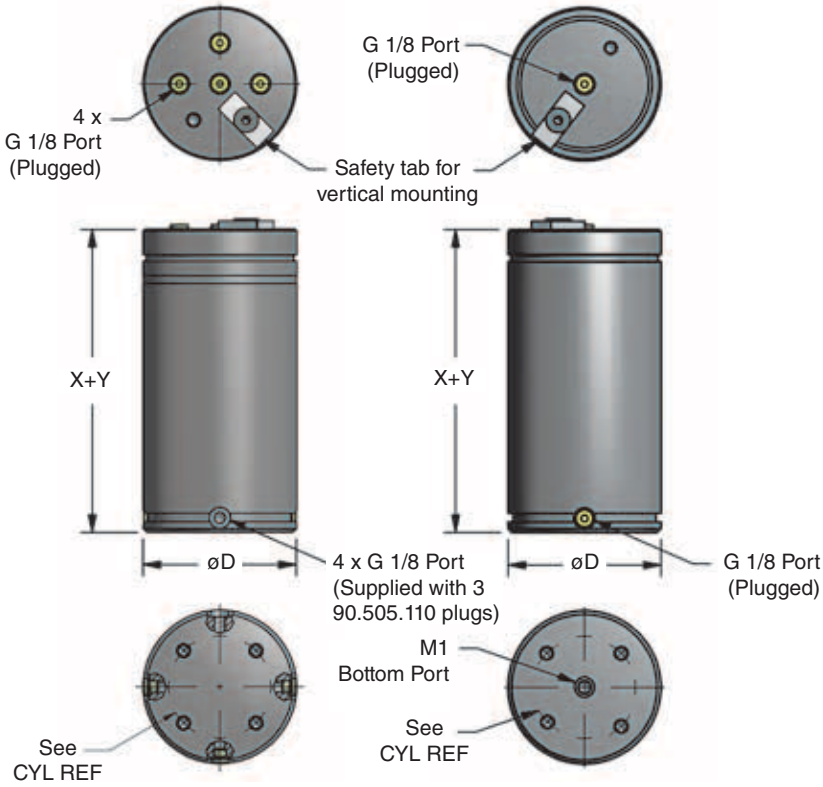
734.207.1100 • 800.DADCO.USA • fax 734.207.2222 • www.dadco.net

Surge Tanks

DADCO surge tanks are used with open-flow systems to increase the volume in the system thereby reducing the pressure rise when cylinders are stroked. The Surge Tank is offered in two Models: F – Free Flow Model has multiple open ports supplied as standard for maximum flexibility when piping; M1– SMS-i® Model has a bottom port to attach to a base plate. Gauges and shut-off ball valves are available upon request.


ST.50.150.B29

For assistance in determining appropriate surge tank size for your system, contact DADCO with the cylinder size, length of stroke being used and amount of pressure rise desired. *DADCO 90.700 (Y-700) / 90.705 (Y-705) hose is generally not recommended for use with surge tanks due to restricted flow capability.*

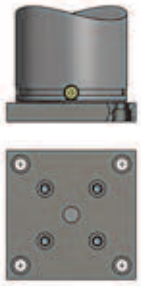


F – Free Flow Model **M1 – SMS-i® Model**

ST	30	50	75	100
D	95 *3.74*	120 *4.72*	150 *5.91*	195 *7.67*
X	117 *4.61*	137 *5.39*	152 *5.98*	157 *6.18*
Y	Volume of Tank L *(in³)*			
50 *1.97*	0.59 *36*	1.05 *64*	1.71 *105*	2.92 *178*
100 *3.94*	0.85 *52*	1.44 *88*	2.33 *142*	3.99 *244*
150 *5.91*	1.10 *67*	1.83 *112*	2.94 *180*	5.06 *309*
200 *7.87*	1.35 *82*	2.22 *136*	3.56 *217*	6.13 *374*
250 *9.84*	1.60 *98*	2.62 *160*	4.17 *254*	7.20 *439*
300 *11.81*	1.85 *113*	3.01 *184*	4.78 *292*	8.27 *505*
350 *13.78*	2.10 *128*	3.40 *208*	5.40 *329*	9.34 *570*
400 *15.74*	2.35 *144*	3.79 *232*	6.01 *367*	10.41 *635*

Preferred Mounts for Surge Tanks.
See pages 11-17 for mount details.

B11 **90.11**

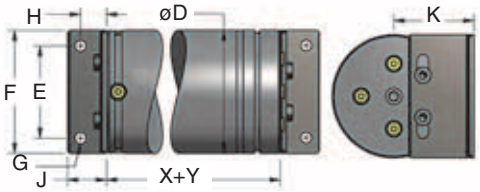


NOTE: B11 mount available with ST.30-ST.75 models only

B21 **90.21**



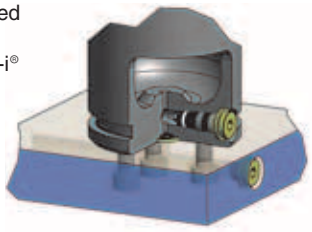
B29 **90.29**



Surge Tank	CYL REF	D	E	F	G	H	J	K
30	3000	95 *3.74*	50 *1.97*	75 *2.95*	4 x M10 *3/8*	25.4 *1.00*	38 *1.50*	50.5 *1.99*
50	5000	120 *4.72*	90 *3.54*	120 *4.72*	4 x M10 *3/8*	25.4 *1.00*	38 *1.50*	78 *3.07*
75	7500	150 *5.91*	90 *3.54*	120 *4.72*	4 x M10 *3/8*	25.4 *1.00*	38 *1.50*	85 *3.35*
100	10000	195 *7.67*	100 *3.94*	150 *5.91*	4 x M12 *1/2*	31.8 *1.25*	50.8 *2.00*	98.5 *3.88*

SMS-i® Surge Tank Connection

DADCO surge tanks ordered with the M1 operating system are used in a SMS-i® and have a bottom port. These tanks are attached to the base plate with a sealing washer and standard mounting hardware.



Ordering Example: ST.30. 150. TO. F

Size:
30, 50, 75, 100

Length (Y):
50,100,150, 200, 250, 300, 400

Operating System:
F = Free Flow Fitting,
M1 = SMS-i® *(Bottom port + sealing component)*

Mount Option:
TO = Basic Model. *When not specified, default is TO.* Mount ordered with cylinder will be attached at factory.

Repair Tools and Accessories

Cartridge Starter Kit
90.335._____
(00500, 00750, 01500, 03000, 05000, 07500, 10000)

The Cartridge Starter Kit includes an Assembly Cap (90.330.__) and an Assembly Cone (90.331.__). The Assembly Cone is used to start the cartridge assembly onto the rod without damaging the seal, the Assembly Cap is used to set the cartridge at a proper depth for C-Ring installation.



90.330.___



90.331.___

Removal Sleeve
90.340._____
(00500, 00750, 01500, 03000, 05000, 07500)

To position the cartridge below the C-ring groove when assembling or disassembling a gas spring.



C-Ring Installation Tool
90.351.00500 (00500 Model)
90.350.00750 (00750 Model)

To insert the C-Style Retaining Ring into the retaining ring groove of 500 and 750 Models.



C-Ring Installation Tool
90.352 (01500 – 07500 Models)
90.352.10000 (10000 Model)

To insert the C-Style Retaining Ring into the retaining ring groove of 01500 – 10000 Models. For more information request bulletin B01101C.



C-Ring Removal Tool
90.355 (00500 – 03000 Models)
90.356 (00750 – 10000 Models)

To remove the C-Style Retaining Ring safely in a single controlled motion.



90.355
Patented



90.356
Patented

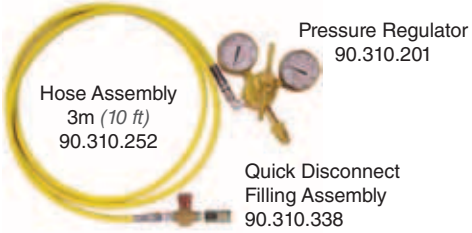
T-Handle
90.320.2 – M8 Thread
90.320.10 – M10 Thread

To remove the piston rod when disassembling and position correctly when reassembling.



Quick Disconnect Charging Assembly
90.310.040

Use the DADCO Quick Disconnect Charging Assembly, *90.310.040*, with the *90.310.111* Charging Nipple or *90.315.5* Pressure Analyzer to charge self-contained gas springs. The *90.310.040* can also be used with a DADCO control panel to charge linked systems. The *90.310.040* includes the *90.310.201* Pressure Regulator, *90.310.252* Hose Assembly and the *90.310.338* Quick Disconnect Filling Assembly.



Quick Disconnect Charging Nipple
90.310.111

Use the DADCO Quick Disconnect Charging Nipple to charge the 90.10 and 90.8 Series gas springs. For more information, contact DADCO.



734.207.1100 • 800.DADCO.USA • fax 734.207.2222 • www.dadco.net

Repair Tools and Accessories

Standard Load Cell
90.300._____
(00500, 00750, 01500, 03000, 05000, 07500, 10000)

The Standard Load Cell gives precise measurement of gas spring charging pressure. Each model requires its specified load cell. Load cells for 00500 – 07500 Models may be used with the Portable Test Stand; the load cell for the 90.10.10000 may be used with an arbor press.



Digital Load Cell
90.305.BGA (Meter)
90.305.LC.05A (22.2 kN Load Cell)
90.305.LC.50A (222 kN Load Cell)

The 90.305.BGA meter can display force in N, Kg or lbs. When paired with the 90.305.LC.05A Load Cell it may be used to measure gas spring force up to 5,000 lbs. When paired with the 90.305.LC.50A Load Cell gas spring force can be measured up to 50,000 lbs. For more information request bulletin B04106C.



DADCO Pressure Analyzer
90.315.5

DADCO's Pressure Analyzer includes two interchangeable bits to easily charge, discharge and gauge the pressure in any DADCO Nitrogen Gas Spring. For more information request bulletin B01133D.



Portable Test Stand
90.305.3

Use the Portable Test Stand in conjunction with a Standard Load Cell for precise measurement of gas spring force. Excludes use with 90.10.10000. For more information request bulletin 97B121.



DADCO Electronic Pressure Monitors

DADCO's Electrical Pressure Monitors indicate if pressure drops below a preset level, alerting the press controller to shut down the press. They are available in a variety of configurations to suit a variety of applications. For more information request bulletin B10105A or contact DADCO.



Valve Bleed Tool
90.360.4

Use the DADCO Valve Bleed Tool to slowly discharge a spring to the desired pressure. For more information contact DADCO.



Port Servicing Tool
90.320.8

To perform all necessary servicing to the valve compartment.



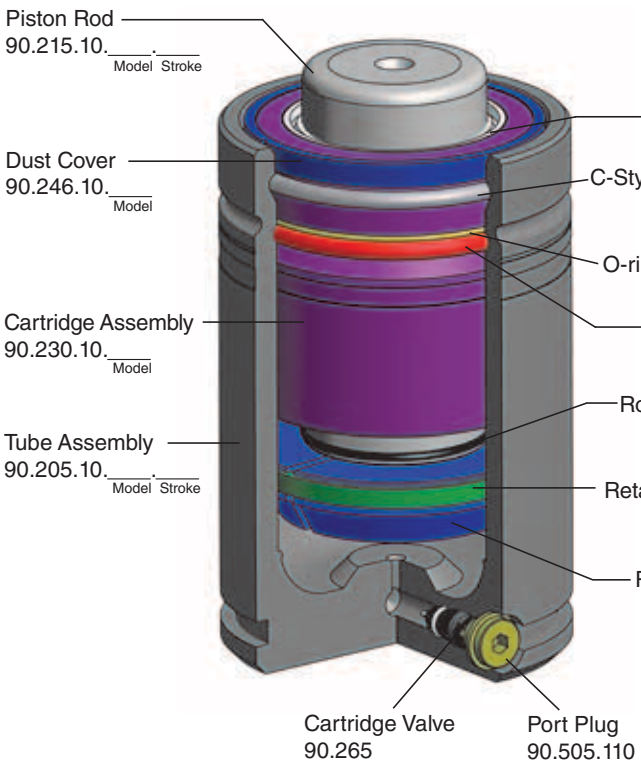
DADCO Repair Tool Kits

DADCO's most popular repair tools are now available in boxed sets for each of our three main classifications of nitrogen gas springs: Micro, Mini and Large Series. Kits may include DADCO load cells, charging nipples and various repair tools. Separate repair tool and load cell sets are also available. For more information request bulletin B05143A.

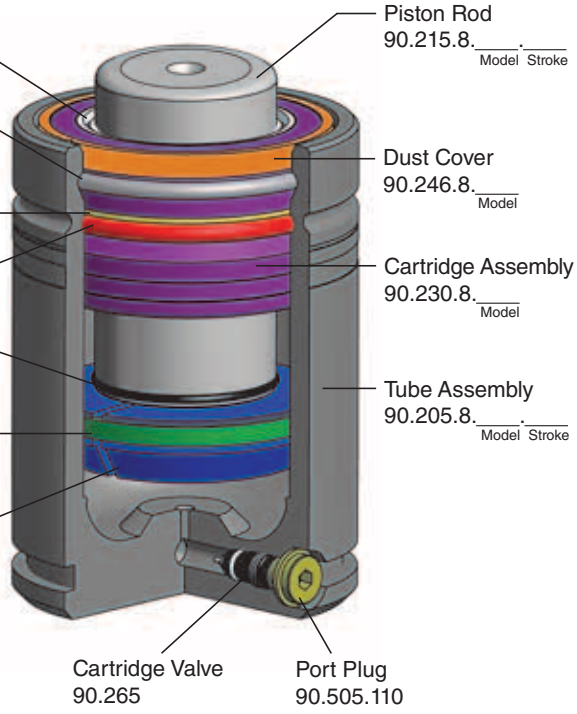


Parts List

90.10 Series Parts List



90.8 Series Parts List



Replacement Part Ordering Example:

Piston Rod: 90.215. 10. *00750.* *025*

Part Number
Series:
90.10 = 10 or 90.8 = 8

Stroke (mm)
Model: 00500, 00750, 01500, 03000, 05000, 07500, 10000
(00500 and 10000 models available in 90.10 Series only)

NOTE: *DADCO's 90.10 and 90.8 Series Nitrogen Gas Springs are permanently marked with model number, serial number and repair kit number. Please refer to these when ordering replacement parts.*

Repair Kits

Include a fully assembled cartridge, dust cover, bottle of assembly oil and a maintenance manual.

90.10 Series	
Model	**Repair Kit No.**
90.10.00500	90.201A.00500
90.10.00750	90.201.00750
90.10.01500	90.201.01500
90.10.03000	90.201.03000
90.10.05000	90.201.05000
90.10.07500	90.201.07500
90.10.10000	90.201.10000

90.8 Series	
Model	**Repair Kit No.**
90.8.00750	90.208.00750
90.8.01500	90.208.01500
90.8.03000	90.208.03000
90.8.05000	90.208.05000
90.8.07500	90.208.07500



734.207.1100 • 800.DADCO.USA • fax 734.207.2222 • www.dadco.net

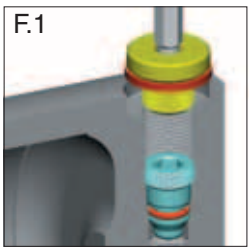
CAUTION

DO NOT attempt maintenance on spring until internal pressure is exhausted.

Operating Specifications

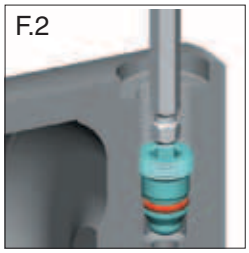
Charging Medium:	Nitrogen Gas
Charging Pressure Range:	15 – 150 bar (*220 – 2175 psi*)
Operating Temperature:	-6°C – 71°C (*20°F – 160°F*)*
Maximum Speed:	1.6 m/sec (*41 in/sec*)

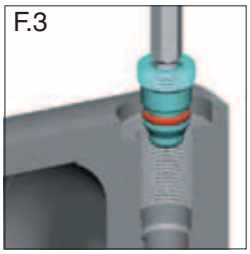
**For extended range applications contact DADCO.*

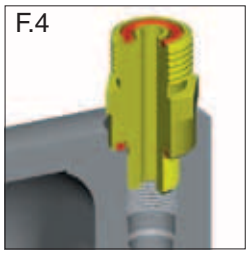


Self-contained to Linked Operation

- Always wear safety glasses when maintaining gas springs. When exhausting pressure, place the gas spring horizontally with the port up for safety.
- Remove port plug, *90.505.110* (F.1).
- With the cylinder in the horizontal position, depress the valve stem using the appropriate tool (F.2). Cover the port with a cloth to absorb discharge.
- After all of the gas pressure is exhausted, be sure that the piston rod will retract into the tube manually. If not, try depressing the valve again. If still unsuccessful, stop and contact your DADCO Service Representative.
- Remove the valve using the appropriate tool (F.3). Lubricate the threads and o-ring of the port adapter being installed (F.4).
- Install port adapter in open port (F.4). A wide variety of port adapters are available, refer to DADCO's Nitrogen Gas Spring Linked System Components Catalog.

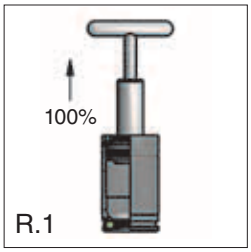


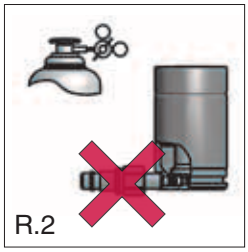




Recharging Self-contained Gas Spring

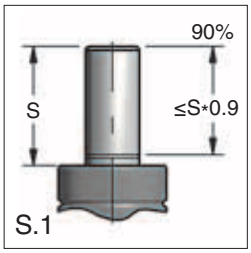
- Hold the spring vertically at all times during filling. Never compress the gas spring in a vice or clamp outside of the die or application as damage to the gas spring can result.
- Never fill a gas spring unless the rod is in the fully extended position (R.1). Filling a gas spring with its rod down can result in improper retaining ring seating (R.2). Thread the T-handle *90.320.2 or 90.320.10* into the rod end and depress the valve stem with the appropriate tool, then pull the rod cartridge assembly up until it is seated firmly against the retaining ring (R.1).
- Charge the gas spring to the desired pressure. Refer to DADCO's 90.8/90.10 Series Maintenance Instructions B06124D for complete step-by-step instruction.

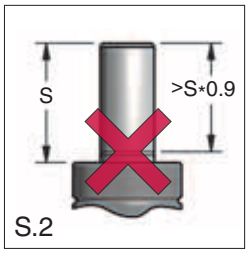




Provide Stroke Reserve

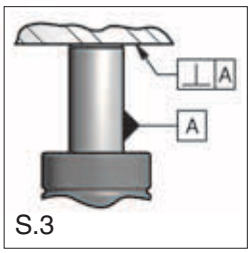
- DADCO's 90.8 and 90.10 Series Gas Springs will permit travel of the full nominal stroke; however, at least a 10% stroke reserve is recommended to achieve optimal performance and safety (S.1, S.2).
- Overstroking the rod or impacting the top of the gas spring will cause permanent damage.

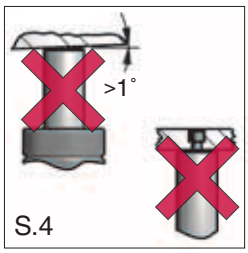




Avoid Side Loading

- Side loading resulting from press action or die construction causes increased wear on bearing, seal and piston rod (S.4). Therefore, avoid side loading when possible (S.3).



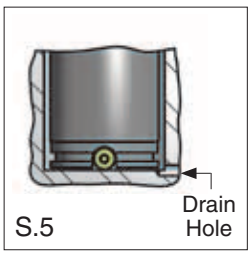


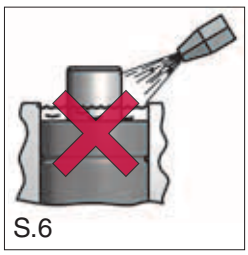
Rod End Thread

- The end of the piston rod has a construction thread intended for assembly and disassembly only and should never be used to mount or secure the gas spring (S.4). Die vibration and/or misalignment will damage the spring.

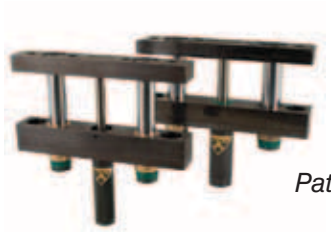
Protect From Fluids

- Direct contact with certain die lubricants and cleaners should be avoided (S.6). Protect gas springs by providing adequate drainage in gas spring pockets (S.5).





Other DADCO Products



Patented

SL2.090 and SL2.180 – Nitrogen Gas Spring Two Post Lifters

- 160 mm and 180 mm rail widths
- Powered by the **Micro 90®** and **Micro 180®**
- Stroke lengths available from 23 mm to 198 mm
- Two post for rail lift applications
- Compact Rail Plate available



SLN.090 and SLN.180 – Micro Nitrogen Gas Lifters

- Non-rotating design; provides lift and guidance
- Two guide rod options for single point, multipoint or rail lift applications
- Compact design powered by the **Micro 90®** and **Micro 180®**
- Stroke lengths available from 25 mm to 125 mm



SCR Series

- From 32 mm to 63 mm in diameter
- Forces up to 28.6 kN
- Full range of standard stroke lengths up to 80 mm
- Completely serviceable



Mini – LJ and L Series

- 38 mm, 45 mm and 50 mm diameters
- Force models: 3 kN, 5 kN, and 7.5 kN
- Full range of standard stroke lengths up to 125 mm



Ultra Force® – U Series

- 19 mm to 195 mm in diameter
- Forces up to 199 kN
- Full range of standard stroke lengths up to 125 mm
- **UltraPak®** cartridge for long life



Patented

Ultra Force Extended® – UX Series

- 25-55% more force on-contact than ISO Standard Nitrogen Gas Springs
- From 50 mm to 195 mm in diameter
- Up to 199 kN of force on contact
- Matches ISO Envelope

DADCO®
43850 Plymouth Oaks Blvd. • Plymouth, MI • 48170 • USA
734.207.1100 • 800.DADCO.USA • Fax 734.207.2222 • www.dadco.net

The global leader in nitrogen gas spring technology

Printed in USA

©DADCO, Inc. 2014 All Rights Reserved

Product changes may occur during the life of this catalog without prior notice, but products supplied will remain functionally interchangeable.